The Law of Portfolio Learnability

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I. Introduction

In the Markowitz (1952) framework, first the portfolio manager estimate the two moments of returns, secondly the PM estimate the portoflio wieghts optimising a criterion.

A different view emerged with the direct portfolio approach of Brandt (1999) and Brandt et al. (2009). Investors ultimately care about portfolio weights, not forecasts of moments. Rather than estimating conditional expected returns and covariance matrices as intermediate objects, one can parameterize and estimate the portfolio policy directly. This changed the object of estimation. This was a very futuristic approach, anticipating of years what computer science and machine learning is doing now.

This estimation problem becomes particularly acute in high-dimensional finance. Kozak et al. (2020) show that the estimation becomes unstable in low-eigenvalue directions. Where the eigenvalues tend to zero, the payoff of the eigenportfolio tend to infinite, creating an arbitrage opportunity. This motivate the economically shrinkage.

More recently, the machine-learning literature has brought model complexity to the foreground. Empirical asset pricing means model with data the financial market. There is always a model behind every modelization. Thus finance can't ignore what is happening in the machine leanring world. Indeed Kelly et al. (2024) and Kelly and Malamud (2025) show that increasing model complexity can improve out-of-sample return prediction even deep into the overparameterized regime. Their conclusion is deliberately strong:

> *"The analyst should always use the largest approximating model that she can compute."*

This was strongly new for the finance world in which the only virute was the parsimony.

This "virtue of complexity" shows that overparameterization can improve approximation and generate useful implicit regularization. Nagel (2025), in turn, shows why the economic meaning of highly overparameterized estimators must be interpreted carefully: large random-feature models

can converge to kernel rules whose behavior is driven by particular forms of historical weighting rather than by an arbitrarily rich set of distinct economic decisions, introducing nominally and the effective complexity.

The overparatized model not only now are in returns prediction, but appears in directly portfolio choice. Simon et al. (2026) replace the linear parametric portfolio policy with deep neural networks and show that flexible nonlinear policies can generate economically valuable allocations, while investor preferences themselves act as a form of economic regularization.

Taken together, this literature establishes that rich representations can be useful and that regularization is if not indispensable, at least usefull.

Yet several fundamental questions remain open.

Building on the direct portfolio approach of Brandt: How reliably, and how quickly, can a flexible and "complex" portfolio rule recover the population-optimal investment decision?
Building on the spectral shrinkage argument of Kozak, Nagel, and Santosh: Which investment directions should be shrunk because they are only weakly identified by returns?
Building on the virtue-of-complexity literature: When is a richer representation a virtue for portfolio choice, and what notion of complexity is economically relevant for the investment decision itself?

Answering these questions requires starting from a more fundamental one:

How much of the optimal portfolio can be learned from a finite market history?

Answering this question requires establishing whether and how fast the portfolio decision can be learned directly, identifying which investment directions should be shrunk because they are only weakly revealed by realized returns, and determining what notion of complexity is economically relevant for portfolio choice—and how much of that complexity a finite market history can support.

First we used a particular estimator (Kernel ridge regression) to quantify our results, later we generalize these to every estimator. Our first contribution is to quantify how rapidly direct portfolio estimation recovers the population tangency direction as market history accumulates. We prove that the directly weights estimation for maximizing the Sharpe ratio is consistent and we characterize the rate at which its estimated direction converges to the population Sharpe. We find that portfolio learnability is governed by the spectrum of managed portfolios. Directions associated with large eigenvalues correspond to investment opportunities that generate substantial payoff variation and are therefore well identified from the available return history. By contrast, low-eigenvalue directions are only weakly identified, so taking large positions along them makes the portfolio especially sensitive to estimation error. More available portfolio directions therefore do not necessarily translate into more usable investment opportunities: what matters is how much of the opportunity set is concentrated in directions that can be estimated with sufficient precision. The result therefore provides both a formal justification for direct portfolio estimation and a quantitative answer to how much market history is required to learn the optimal investment decision.
Our first contribution implies even a quantification of the optimal shrinkage: It should decline as market history accumulates and as a function of the managed portfolio spectrum. Additional observations progressively reveal portfolio directions that were previously too weakly identified to support reliable investment positions, allowing the investor to exploit a broader set of economically meaningful opportunities.

Our second contribution is what we call the law of effective portfolio complexity. We distinguish two notions of complexity that play fundamentally different economic roles. The first is nominal representation complexity: the size of the approximating representation, as measured, for example, by the number of features, basis functions, or parameters. Kelly, Malamud, and Zhou show that, under their proportional high-dimensional asymptotic framework, increasing nominal representation size can improve limiting out-of-sample return-prediction performance. Richer representations can therefore be valuable because they expand approximation capacity and the set of investment rules that can potentially be expressed.

The second notion is effective portfolio complexity, which we define directly in the space of the investment decision. It measures the effective breadth of the portfolio: the effective number of economically distinct managed-portfolio directions that remain active after shrinkage. This quantity need not move one-for-one with representation size. A representation may be arbitrarily rich while the portfolio learned from it remains effectively low dimensional. We show that the optimal level of effective portfolio complexity is finite at every finite sample size and increases as additional market history becomes available. Too little effective complexity leaves valuable investment opportunities unused; moving beyond the data-supported scale increasingly activates directions that the available history only weakly identifies. A rich representation may be a virtue; how much of that richness should become an active investment decision is a separate learnability question.

Our third and last contribution is what we call the fundamental limit of portfolio learning. We show that the rate at which the optimal portfolio can be recovered is not merely a property of our estimator. No portfolio-learning procedure —including neural networks, random forests, or alternative kernel methods— can uniformly improve this rate over the same class of economies without imposing additional structure.

The learning limit therefore does not belong to a particular estimator. It belongs to the investment problem itself.

Taken together, our results characterize how rapidly the population tangency portfolio can be recovered from finite market history, how richer representations expand the set of potential investment opportunities, and how the effective portfolio complexity that can be reliably exploited grows as additional data become available. We also establish the fastest rate at which any procedure can achieve this recovery. Our results therefore quantify both the opportunities created by increasingly flexible representations and the fundamental limits that finite market history imposes on machine learning for portfolio choice in finance.

II. Population Problem Portfolio Choice

At date t, the portfolio manager faces N risky stocks. The corresponding next-period excess-return vector is $R_{t+1} = \left(R_{1,t+1}, \ldots, R_{N,t+1}\right)' \in \mathbb{R}^N$. For each stock $i = 1, \ldots, N$, the portfolio manager observes a D-dimensional vector of firm characteristics $Z_{i,t} \in \mathcal{Z} \subseteq \mathbb{R}^D$. The portfolio manager has to choose the portfolio weights policy, between an infinte number of policies in an Hilbert space $\mathcal{H}$, at every time, this maps characteristics into weights: $W \in \mathcal{H} : \mathcal{Z} \to \mathbb{R}$. The resulting portfolio return at each time is

$$R^p_{t+1}(W) = \frac{1}{N} \sum_{i=1}^{N} W(Z_{i,t}) R_{i,t+1}. \tag{1}$$

The factor $1/N$ is a cross-sectional normalization that prevents portfolio scale from growing mechanically with the number of available stocks.

Definition 1 (Economy)**.** An economy ε is a probability law for the stochastic process

$$\{(N, Z_t, R_{t+1})\}_{t \in \mathbb{Z}},$$

defined on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in \mathbb{Z}}, \mathbb{P}_\varepsilon)$. We require Z_t to be $\mathcal{F}_t$-measurable and R_{t+1} to be $\mathcal{F}_{t+1}$-measurable.

For a fixed economy, expectations are taken under $\mathbb{P}_\varepsilon$; we suppress the economy subscript when it is unambiguous. The population criterion is

$$Q(W) = \mathbb{E}\left[\left(1 - R^p_{t+1}(W)\right)^2\right], \qquad W^\star \in \arg\min_{W \in \mathcal{H}} Q(W). \tag{2}$$

The following proposition relates this criterion to the maximum squared Sharpe ratio. All population statements refer to the same one-period state-return distribution.

Proposition 1 (Maximum-Sharpe direction; cf. Britten-Jones (1999))**.** *Suppose every policy in $\mathcal{H}$ generates a square-integrable payoff, the population minimum is attained, and $0 < Q(W^\star) < 1$. Sharpe ratios below are evaluated only for policies with positive payoff variance. Every nonzero policy $W \in \mathcal{H}$ can be written as*

$$W = cU, \qquad c \in \mathbb{R}, \qquad \|U\|_{\mathcal{H}} = 1.$$

The normalized policy U determines the portfolio direction, namely the relative state-dependent allocation rule, whereas c determines the overall scale of exposure along that direction. Then, with the minimization over U restricted to positive-variance payoffs,

$$\min_{W \in \mathcal{H}} Q(W) = \min_{\substack{U \in \mathcal{H} \\ \|U\|_{\mathcal{H}}=1}} \frac{1}{1 + \mathrm{SR}^2\left(R^p_{t+1}(U)\right)},$$

where

$$\mathrm{SR}\left(R^p_{t+1}(U)\right) := \frac{\mathbb{E}\left[R^p_{t+1}(U)\right]}{\sqrt{\mathrm{Var}\left(R^p_{t+1}(U)\right)}}.$$

Hence, minimizing Q selects a maximum-Sharpe direction and fixes its scale, with positive expected excess return.

We embrace the philosophy of Brandt (1999): rather than estimating expected returns and the covariance matrix as intermediate objects and then solving for the optimal portfolio, we estimate portfolio weights directly. Estimating the mean return vector requires estimating N expected returns, while estimating the covariance matrix requires estimating $N(N+1)/2$ distinct covariance parameters. By contrast, directly estimating the portfolio means estimating the N portfolio weights that are ultimately the object of interest.

The criterion in (2) deliberately describes a frictionless portfolio problem. More generally, the loss should represent the economic objective actually faced by the portfolio manager. Transaction costs,

market impact, short-selling costs, leverage constraints, or other implementation frictions should therefore be incorporated directly into the portfolio objective whenever they are economically relevant. Estimating the frictionless-optimal policy and subtracting costs afterwards generally does not deliver the same decision.

Proposition 2 (Pricing kernel and Hansen–Jagannathan frontier)**.** *Let $W_\varepsilon^\star$ denote the population-optimal policy in Proposition 1, and define*

$$m_{t+1,\varepsilon}^\star := 1 - R_{t+1}^p(W_\varepsilon^\star).$$

Then $m_{t+1,\varepsilon}^\star$ prices every managed excess-return portfolio:

$$\mathbb{E}_\varepsilon\left[m_{t+1,\varepsilon}^\star R_{t+1}^p(V)\right] = 0, \qquad \forall V \in \mathcal{H}.$$

Moreover, letting $\mathrm{SR}_\varepsilon^\star := \mathrm{SR}(R_{t+1}^p(W_\varepsilon^\star))$*,*

$$\mathbb{E}_\varepsilon[m_{t+1,\varepsilon}^\star] = \mathbb{E}_\varepsilon[(m_{t+1,\varepsilon}^\star)^2] = \frac{1}{1+(\mathrm{SR}_\varepsilon^\star)^2}.$$

Hence the normalized pricing kernel

$$\widetilde{m}_{t+1,\varepsilon} := \frac{m_{t+1,\varepsilon}^\star}{\mathbb{E}_\varepsilon[m_{t+1,\varepsilon}^\star]}$$

satisfies $\mathbb{E}_\varepsilon[\widetilde{m}_{t+1,\varepsilon}] = 1$*, and* $\mathrm{Var}_\varepsilon(\widetilde{m}_{t+1,\varepsilon}) = (\mathrm{SR}_\varepsilon^\star)^2$*. Equivalently,*

$$\frac{\sigma_\varepsilon(\widetilde{m}_{t+1,\varepsilon})}{\mathbb{E}_\varepsilon[\widetilde{m}_{t+1,\varepsilon}]} = \mathrm{SR}_\varepsilon^\star,$$

so the Hansen–Jagannathan bound ((Hansen and Jagannathan, 1991)) holds with equality.

The proposition provides the asset-pricing dual of Proposition 1. The same portfolio that maximizes the Sharpe ratio generates, through $m_{t+1,\varepsilon}^\star = 1 - R_{t+1}^p(W_\varepsilon^\star)$, the minimum-variance unit-mean

pricing kernel for the managed-payoff space. Thus the maximum-Sharpe portfolio and the Hansen–Jagannathan frontier are two representations of the same population geometry.

III. Direct Portfolio Learning

To estimate the portfolio policy we use a kernel representation Filipović and Schneider (2024); Nagel (2025) because it provides a common framework for portfolio policies of very different forms, as shown in Table IV. Indeed, depending on the kernel, the same estimator can generate static, linear, polynomial or sieve-based, and flexible nonlinear mappings, generalizing respectively the static mean–variance portfolio of Britten-Jones (1999), the linear parametric portfolio policies of Brandt et al. (2009), and the semiparametric spline-based policies of Caldeira et al. (2023).
Kernel methods are also closely connected to the recent work of Nagel (2025); Kelly and Malamud (2025). In particular, random Fourier features (Rahimi and Recht, 2007) provide finite-dimensional approximations to stationary kernels, while the corresponding kernel estimator can be viewed as the infinite-feature limit. In this sense, the kernel formulation provides a natural benchmark for portfolio models whose nominal representation complexity is allowed to become arbitrarily large.

Kernel methods are also closely related to neural-network representations. A random-feature approximation can be interpreted as a neural network with a fixed, randomly generated hidden layer, in which only the final linear coefficients are estimated. More generally, neural tangent kernels arise as the infinite-width limits of particular neural-network training regimes (Jacot et al., 2018).

Let

$$K : \mathscr{Z} \times \mathscr{Z} \to \mathbb{R}$$

be a positive-definite scalar kernel and let $\mathcal{H}_K$ denote the corresponding space of characteristic-based portfolio policies.

Assumption 1 (Portfolio representation)**.** The space $\mathscr{Z}$ is Polish. Fix a jointly measurable kernel K, with separable RKHS $\mathcal{H}_K$, satisfying

$$\sup_{z \in \mathscr{Z}} K(z, z) \leq \kappa_K < \infty.$$

The representation is fixed across economies and sample sizes.

Let $W_K^\star$ denote the minimum-norm population minimizer in $\mathcal{H}_K$, when attained. Write $W_\varepsilon^\star = W_{\varepsilon,K}^\star$ and $\Sigma_{\mathcal{H}_K,\varepsilon}$ when distinguishing economies.

The empirical regularized counterpart of the population problem in Equation (2) is

$$\begin{aligned}\widehat{W}_{\lambda,T} &:= \underset{W\in\mathcal{H}_K}{\arg\min}\left\{\frac{1}{T}\sum_{t=1}^{T}\left(1-\frac{1}{N}\sum_{i=1}^{N}W(Z_{i,t})R_{i,t+1}\right)^2+\lambda\|W\|_{\mathcal{H}_K}^2\right\}\\ &= \underset{W\in\mathcal{H}_K}{\arg\min}\left\{\widehat{Q}(W)+\lambda\|W\|_{\mathcal{H}_K}^2\right\} = \min_{\substack{U\in\mathcal{H}\\ \|U\|_{\mathcal{H}}=1}}\left\{\frac{1}{1+\widehat{\mathrm{SR}}_{\lambda,T}^2\left(R_{t+1}^p(U)\right))}\right\}, \qquad \lambda>0.\end{aligned} \tag{3}$$

where

$$\widehat{\mathrm{SR}}_{\lambda,T}\left(R_{t+1}^p(U)\right) := \frac{\frac{1}{T}\sum_{t=1}^{T}R_{t+1}^p(U)}{\sqrt{\frac{1}{T}\sum_{t=1}^{T}\left(R_{t+1}^p(U)-\frac{1}{T}\sum_{s=1}^{T}R_{s+1}^p(U)\right)^2+\lambda}}.$$

The economic structure of the estimator becomes transparent by viewing kernel-managed portfolio payoffs as elements of the portfolio policy space. At each date t, define

$$X_t := \frac{1}{N}\sum_{i=1}^{N}R_{i,t+1}K(\cdot,Z_{i,t})\in\mathcal{H}_K.$$

Thus X_t is a random element of $\mathcal{H}_K$. If evaluated at $z\in\mathcal{Z}$, it gives

$$X_t(z) = \frac{1}{N}\sum_{i=1}^{N}R_{i,t+1}K(z,Z_{i,t})\in R.$$

the payoff generated by the kernel-managed portfolio that assigns stock i the weight $K(z,Z_{i,t})/N$. Its population mean is

$$\mu_{\mathcal{H}_K} := \mathbb{E}[X_t]\in\mathcal{H}_K,$$

while its population second-moment operator is

$$\Sigma_{\mathcal{H}_K} := \mathbb{E}[X_t \otimes X_t] : \mathcal{H}_K \to \mathcal{H}_K,$$

where, for every $V \in \mathcal{H}_K$,

$$(X_t \otimes X_t)V = \langle V, X_t \rangle_{\mathcal{H}_K} X_t.$$

The first moment captures expected managed-portfolio payoffs, while the second-moment operator describes the magnitude of realized payoffs across portfolio directions.
Their sample counterparts are the element

$$\widehat{\mu}_T = \frac{1}{T} \sum_{t=1}^{T} X_t \in \mathcal{H}_K,$$

and the operator

$$\widehat{\Sigma}_T = \frac{1}{T} \sum_{t=1}^{T} X_t \otimes X_t : \mathcal{H}_K \to \mathcal{H}_K.$$

Proposition 3 (Managed-portfolio Markowitz representation)**.** *The population-optimal minimum-norm policy satisfies*

$$\mu_{\mathcal{H}_K} = \Sigma_{\mathcal{H}_K} W_K^\star. \tag{4}$$

Equivalently,

$$W_K^\star = \Sigma_{\mathcal{H}_K}^\dagger \mu_{\mathcal{H}_K},$$

where $\Sigma_{\mathcal{H}_K}^\dagger$ denotes the pseudo inverse.
The regularized sample policy in Equation (3) *satisfies*

$$\widehat{W}_{\lambda,T} = \left(\widehat{\Sigma}_T + \lambda I\right)^{-1} \widehat{\mu}_T, \qquad \lambda > 0. \tag{5}$$

Equation (5) provides the structural link to classical portfolio choice. The investor is still com-

bining expected returns against second moments, as in Markowitz. The difference is that the investment objects are now characteristic-managed portfolios weighted with the kernels.

Regularization has a familiar portfolio interpretation. As emphasized by Kozak et al. (2020), portfolio directions with very little payoff variation but non-negligible expected returns resemble near-arbitrage opportunities and should therefore be treated with particular caution. Regularization has a familiar portfolio interpretation. As emphasized by Kozak et al. (2020), weak portfolio directions are particularly dangerous because small estimation errors can translate into very large portfolio positions. In their eigenportfolio setting, a low-variance direction with a sufficiently large estimated expected return can even resemble a near-arbitrage opportunity.

The same instability appears in our managed-portfolio representation. Let $\Sigma_{\mathcal{H}_K} e_j = \mu_j e_j$, and let $m_j := \langle \mu_{\mathcal{H}_K}, e_j \rangle_{\mathcal{H}_K}$. Without regularization, the portfolio weight along direction e_j is proportional to

$$\widehat{W}_{0,j} = \frac{\widehat{m}_j}{\widehat{\mu}_j}.$$

Hence, when $\widehat{\mu}_j$ is small, even modest estimation error in $\widehat{m}_j$ can generate a very large portfolio position. Regularization replaces the small eigenvalue in the denominator by $\widehat{\mu}_j + \lambda$:

$$\widehat{W}_{\lambda,j} = \frac{\widehat{m}_j}{\widehat{\mu}_j + \lambda} = \frac{\widehat{\mu}_j}{\widehat{\mu}_j + \lambda} \widehat{W}_{0,j}.$$

Thus, $\widehat{\mu}_j/(\widehat{\mu}_j + \lambda)$ is the spectral shrinkage factor.

The spectral filter derived above suggests a natural measure of how much of the managed-portfolio opportunity set is effectively used by the investor. The relevant notion of portfolio complexity is not the number of characteristics, parameters, or random features in the representation, but the number of economically distinct portfolio directions that remain active after shrinkage.

This motivates the following definition.

Definition 2 (Effective portfolio complexity)**.** For a fixed portfolio representation K and $\lambda > 0$,

define

$$C_K(\lambda) := \mathrm{Tr}\left[\Sigma_{\mathcal{H}_K}(\Sigma_{\mathcal{H}_K} + \lambda I)^{-1}\right] = \sum_{j\geq 1} \frac{\mu_j}{\mu_j + \lambda}. \tag{6}$$

Effective portfolio complexity is the shrinkage-adjusted number of managed-portfolio directions used by the portfolio rule.

The definition follows directly from the spectral shrinkage mechanism already introduced. Each managed-portfolio direction contributes between zero and one to $C_K(\lambda)$. If $\mu_j \gg \lambda$, its shrinkage factor is close to one and the direction is effectively active. If $\mu_j \ll \lambda$, the direction contributes little and is effectively suppressed.

Effective complexity therefore measures how deeply into the managed-portfolio spectrum the investor is actually operating.

This notion is fundamentally different from nominal representation size. A model may contain thousands, or infinitely many, basis directions while the portfolio decision itself remains effectively low dimensional because regularization suppresses most weak directions. Conversely, a rich representation can be valuable because it makes additional investment opportunities available once sufficient market history exists to identify them.

IV. Portfolio Learnability

A. A Class of Portfolio Economies

For the fixed representation in Assumption 1, fix $b > 1$, $\gamma > b$, constants $C_\beta, B_X, R > 0$, $K_4 \geq 1$, and $0 < c_\mu \leq C_\mu < \infty$. We denote by

$$\mathcal{E} = \mathcal{E}(b, \gamma, C_\beta, B_X, K_4, R, c_\mu, C_\mu)$$

the nonempty class of economies ε satisfying the following conditions, with all bounds common to the class:

(E1) The process $\{(Z_t, R_{t+1})\}_{t\in\mathbb{Z}}$ is strictly stationary and polynomially β-mixing, with $\beta_\varepsilon(k) \leq C_\beta k^{-\gamma}$ for every $k \geq 1$.

(E2) *Gaussian-type managed-return tails and conditional fourth moments.* Let

$$K_t := \left[K(Z_{i,t}, Z_{j,t})\right]_{i,j=1}^{N}.$$

Assume

$$\mathbb{E}_\varepsilon\left[\exp\left(\frac{N^{-2}R'_{t+1}K_tR_{t+1}}{B_X^2}\right)\middle|\mathscr{F}_t\right] \leq 2 \quad \text{a.s.} \tag{7}$$

For every $V \in \mathcal{H}_K$, assume also

$$\mathbb{E}_\varepsilon\left[(R^p_{t+1}(V))^4\middle|\mathscr{F}_t\right] \leq K_4\left\{\mathbb{E}_\varepsilon\left[(R^p_{t+1}(V))^2\middle|\mathscr{F}_t\right]\right\}^2 \quad \text{a.s.} \tag{8}$$

Both bounds are uniform over the class. Since $N^{-2}R'_{t+1}K_tR_{t+1} = \|X_t\|^2_{\mathcal{H}_K}$, Condition (7) directly controls the diagonal of the induced kernel while allowing unbounded returns. Conditional Gaussian returns with uniformly bounded conditional second moments satisfy Condition (E2); see Proposition 4.

(E3) The population minimum over $\mathcal{H}_K$ is attained. Its minimum-norm solution $W_\varepsilon^\star$ satisfies $\|W_\varepsilon^\star\|_{\mathcal{H}_K} \le R$.

(E4) The pricing kernel $m_{t+1,\varepsilon}^\star := 1 - R_{t+1}^p(W_\varepsilon^\star)$ satisfies

$$\mathbb{E}_\varepsilon\left[m_{t+1,\varepsilon}^\star R_{t+1}^p(V) \mid \mathscr{F}_t\right] = 0, \qquad \forall\, V \in \mathcal{H}_K, \quad \text{a.s.}$$

Here $\mathscr{F}_t$ is the investor's information at the decision date in Definition 1. Conditional pricing is imposed separately from unconditional population optimality; positivity is not imposed.

(E5) Let $\Sigma_{\mathcal{H}_K,\varepsilon} = \mathbb{E}_\varepsilon[X_t \otimes X_t]$ and let $\mu_{1,\varepsilon} \ge \mu_{2,\varepsilon} \ge \cdots > 0$ denote its positive eigenvalues. Then

$$c_\mu j^{-b} \le \mu_{j,\varepsilon} \le C_\mu j^{-b}, \qquad j \ge 1.$$

(E6) The optimal payoff has positive variance and $\mathrm{SR}_{\varepsilon,K}^\star := \mathrm{SR}(R_{t+1}^p(W_\varepsilon^\star))$ satisfies the common bounds

$$0 < \underline{S} \le \mathrm{SR}_{\varepsilon,K}^\star \le \overline{S} < \infty.$$

Condition (E2) replaces the almost-sure bounded-return restriction by Gaussian-compatible tail and fourth-moment control. Proposition 4 in Appendix A.4 gives a simple sufficient conditional-Gaussian specification. No Gaussianity is otherwise imposed.

The class $\mathcal{E}$ separates two forces governing portfolio learnability. The exponent γ controls the persistence of the data, while b controls the geometry of the managed-portfolio spectrum. The parameter b describes how concentrated the managed-portfolio opportunity set is. A large b means that most payoff variation is concentrated in a relatively small number of strong portfolio directions. A smaller b leaves a longer tail of economically distinct but progressively weaker directions.

Polynomial spectral decay is natural for finite-smoothness representations, including Sobolev–Matérn, Laplace, and spline-type kernels. Gaussian kernels can instead generate exponential or faster-than-polynomial decay (Belkin, 2018). For exposition, we use $b = \infty$ as shorthand for the

fast-decay benchmark, including Gaussian-type spectra and the finite-rank limit. The class $\mathcal{E}$ itself has finite b; these are separate benchmarks, not substitutions into the restriction $\gamma > b$. Their relevance depends on the spectrum of $\Sigma_{\mathcal{H}_K,\varepsilon}$, not on the kernel name alone.

B. Learnable Sharpe ratio and optimal shrinkage

We can now ask how regularization should change as market history accumulates, how rapidly the population maximum-Sharpe direction can be recovered, and how much of the managed-portfolio opportunity set can be reliably used. The financial observations can be independent over time or weakly dependent.

Population loss and Sharpe of an estimated policy hold the training sample fixed and integrate over the stationary one-period evaluation law. They are not the Sharpe ratio of a finite test sample.

The statistical upper bound follows from Bignozzi et al. (2026, Lemma A.7). Appendix A.5 verifies the kernel representation and its statistical assumptions.

Theorem 1 (Optimal shrinkage and learnable frontier)**.** *Choose the optimal regularization*

$$\lambda_T^\star \asymp T^{-b/(b+1)}$$

, with comparison constants independent of the economy. Uniformly over $\varepsilon \in \mathcal{E}$,

$$Q_\varepsilon(\widehat{W}_{\lambda_T^\star,T}) - Q_\varepsilon(W_\varepsilon^\star) = O_{\mathbb{P}_\varepsilon}\left(T^{-b/(b+1)}\right). \tag{9}$$

Moreover, uniformly over $\varepsilon \in \mathcal{E}$,

$$\mathrm{SR}_{\varepsilon,K}^\star - \mathrm{SR}\left(R_{t+1}^p(\widehat{W}_{\lambda_T^\star,T})\right) = O_{\mathbb{P}_\varepsilon}\left(T^{-b/(b+1)}\right), \tag{10}$$

where:

$$SR_{\varepsilon}\left(\widehat{W}_{K,\lambda,T}\right) = \frac{\mathbb{E}_{\varepsilon}\left[R^{p}_{t+1}\left(\widehat{W}_{K,\lambda,T}\right)\middle|\widehat{W}_{K,\lambda,T}\right]}{\sqrt{\mathrm{Var}_{\varepsilon}\left(R^{p}_{t+1}\left(\widehat{W}_{K,\lambda,T}\right)\middle|\widehat{W}_{K,\lambda,T}\right)}} = \frac{\mathbb{E}_{\varepsilon}\left[\frac{1}{N}\sum_{i=1}^{N}\widehat{W}_{K,\lambda,T}(Z_{i,t})R_{i,t+1}\middle|\widehat{W}_{K,\lambda,T}\right]}{\sqrt{\mathrm{Var}_{\varepsilon}\left(\frac{1}{N}\sum_{i=1}^{N}\widehat{W}_{K,\lambda,T}(Z_{i,t})R_{i,t+1}\middle|\widehat{W}_{K,\lambda,T}\right)}}$$

and

$$SR^{\star}_{\varepsilon,K} = \sup_{W\in\mathcal{H}_K}\frac{\mathbb{E}_{\varepsilon}\left[\frac{1}{N}\sum_{i=1}^{N}W(Z_{i,t})R_{i,t+1}\right]}{\sqrt{\mathrm{Var}_{\varepsilon}\left(\frac{1}{N}\sum_{i=1}^{N}W(Z_{i,t})R_{i,t+1}\right)}}.$$

The theorem first establishes consistency of direct portfolio learning. Since $\lambda^{\star}_{T} \to 0$ and $\mathrm{SR}^{\star}_{K} - \mathrm{SR}\left(\widehat{W}_{\lambda^{\star}_{T},T}\right) \to_{P} 0$, the portfolio manager eventually recovers the maximum-Sharpe direction available within the chosen portfolio representation.

The optimal regularization path also gives a dynamic interpretation to the eigenportfolio logic discussed above. With a short return history, directions deep in the managed-portfolio spectrum are weakly identified and are therefore shrunk heavily. As more observations become available, their expected payoff and payoff variation can be estimated more precisely, allowing the manager to take progressively larger positions in those directions.

Indeed, as $\lambda^{\star}_{T} \asymp T^{-b/(b+1)}$ falls with T, the spectral filter $\frac{\mu_j}{\mu_j+\lambda^{\star}_{T}}$ moves toward one for progressively smaller values of μ_j. Directions that were effectively suppressed in short samples therefore become statistically usable in longer samples.

Regularization is thus a finite-history device. More data allow the portfolio manager to move deeper into the managed-portfolio spectrum and exploit investment opportunities that were previously too weakly identified.

The shape of the spectrum determines how quickly portfolio opportunities can be learned. A steep spectrum (large b) concentrates economically relevant payoff variation in a few directions, making the portfolio easier to learn. A flatter spectrum spreads variation across many weaker directions, increasing potential breadth but slowing learning.

Importantly, portfolio learnability is therefore not determined directly by the number D of firm characteristics. Two representations with the same number of characteristics can generate very different learning problems. If their characteristics produce largely redundant portfolio bets, the managed-portfolio spectrum is concentrated. If instead they produce many distinct but weak portfolio directions, the spectrum is diffuse and substantially more market history is required.

What matters for the portfolio manager is therefore not the raw number of signals, but the economic geometry of the portfolios generated by those signals.

Figure 1 gives a graphical interpretation of the learning result: as market history grows, the learned capital-allocation line rotates toward the population tangency line. Under Condition (E6), at any fixed volatility, the expected-return gap closes at rate $O_P\left(T^{-b/(b+1)}\right)$.

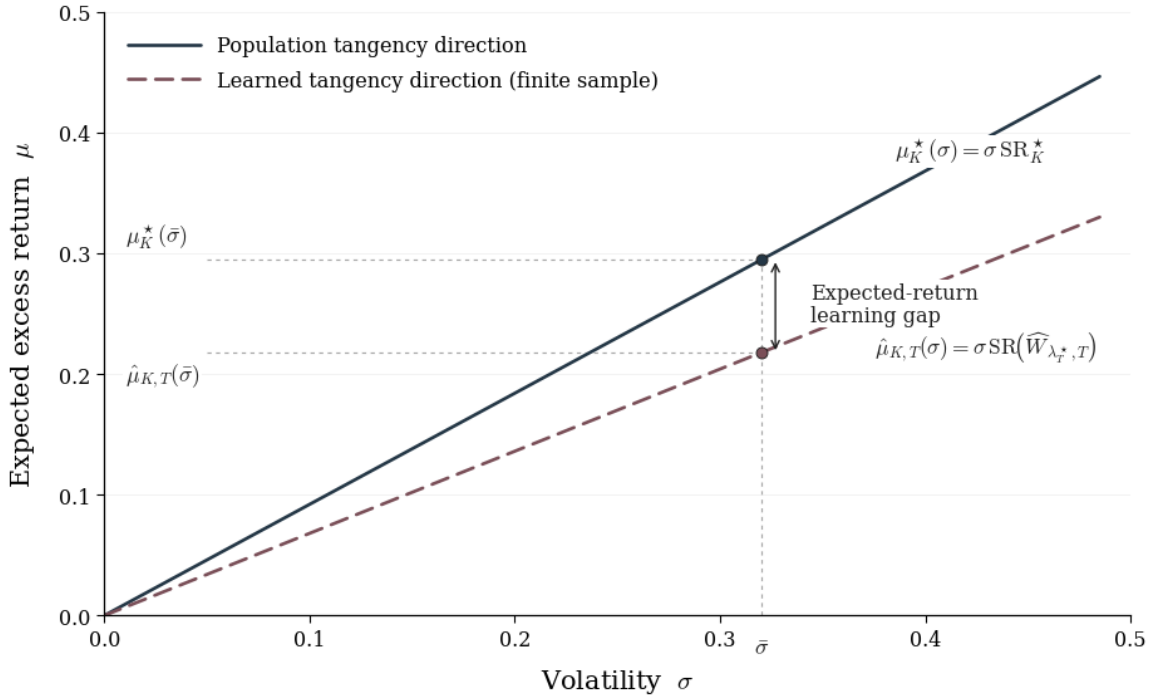


Figure 1. Learning the tangency direction. The population tangency direction generates expected excess return $\mu_K^\star(\sigma) = \sigma\,\mathrm{SR}_K^\star$, while the direction learned from a finite history generates $\hat\mu_{K,T}(\sigma) = \sigma\,\mathrm{SR}(\widehat{W}_{\lambda_T^\star,T})$. At the fixed volatility target $\bar\sigma$, the vertical distance is the expected-return learning gap $\mu_K^\star(\bar\sigma) - \hat\mu_{K,T}(\bar\sigma)$.

C. Optimal effective portfolio complexity

The optimal-shrinkage result can now be expressed directly in terms of portfolio complexity.

Theorem 2 (Learnable effective portfolio complexity)**.** *Under the portfolio-loss conditions of Theorem 1, the rate-optimal effective complexity satisfies*

$$C_{K,T}^\star := C_K(\lambda_T^\star) \asymp T^{1/(b+1)}. \tag{11}$$

The theorem gives a direct economic interpretation of additional market history. As T increases and optimal shrinkage declines, the portfolio manager can reliably activate a broader set of managed-portfolio directions.

This expansion is nevertheless gradual. Effective complexity grows sublinearly: $\frac{C_{K,T}^\star}{T} \to 0$.

Thus, even when the underlying representation is arbitrarily rich, a finite market history supports only a limited fraction of its potential portfolio directions.

A steep managed-portfolio spectrum implies slower growth of effective complexity because most economically relevant payoff variation is already concentrated in a few strong directions. A flatter spectrum supports greater potential breadth, but exploiting that breadth requires substantially more market history.

Figure 2 summarizes this tradeoff. With a short history, the Sharpe-maximizing portfolio uses only a small number of effective directions. With more observations, weaker directions become reliably investable and the optimal effective complexity shifts outward.

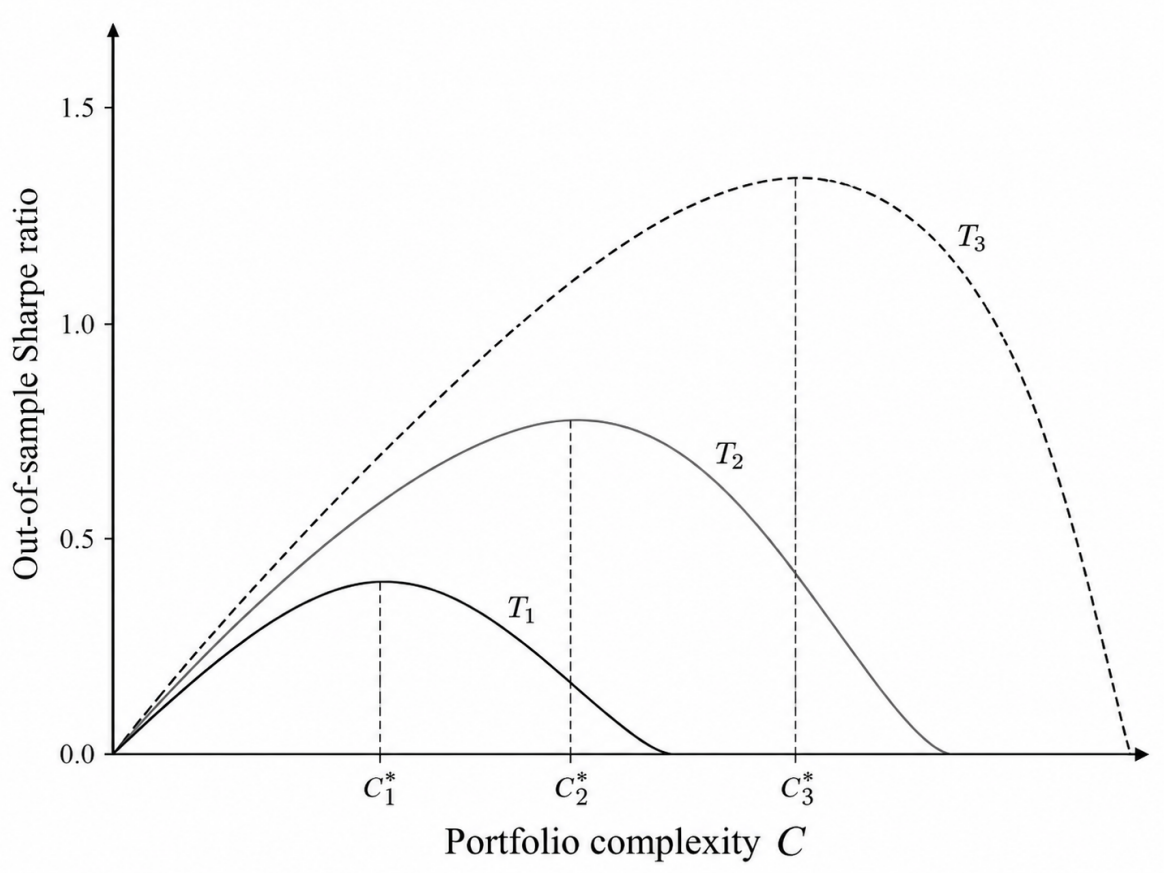


Figure 2. Market history and learnable portfolio complexity. Schematic Sharpe–complexity profiles for different history lengths. Longer market histories allow the portfolio manager to activate a broader set of managed-portfolio directions, shifting the Sharpe-maximizing effective complexity to the right.

The distinction from nominal size is central. A large representation expands the set of portfolio policies that can potentially be expressed. Effective complexity measures how much of that representation the portfolio manager can actually use given the available history of returns. Effective complexity is therefore a learnable economic that quantify the amount of eigenportoflios that the portoflio manager trade.

D. The fundamental limit

The previous results describe how a portfolio manager should use a finite market history: weak managed-portfolio directions should be shrunk, the amount of shrinkage should decline as more observations become available, and the set of portfolio opportunities that can be reliably exploited expands with the sample. A natural question remains. Could a different learning technology overcome this limitation? Could a sufficiently large neural network, a richer random-feature model, or a more sophisticated portfolio algorithm recover the optimal investment decision substantially faster?

The answer is no without additional economic structure.

Theorem 3 (Fundamental limit of maximum-Sharpe learning)**.** *There exists a constant $c > 0$ such that, for all sufficiently large T,*

$$\inf_{\widetilde{W}_T} \sup_{\varepsilon \in \mathcal{E}} \mathbb{E}_\varepsilon \left[\mathrm{SR}^\star_{\varepsilon,K} - \mathrm{SR}_\varepsilon\left(R^p_{t+1}(\widetilde{W}_T) \right) \right] \geq c\, T^{-b/(b+1)}.$$

The infimum is over all measurable portfolio-learning procedures based on the T observed financial rows. Hence no portfolio-learning procedure can uniformly recover the population maximum Sharpe ratio at a faster rate than $T^{-b/(b+1)}$ over $\mathcal{E}$.

Together with Theorem 1, this establishes $T^{-b/(b+1)}$ as the minimax-optimal rate for learning the maximum Sharpe ratio over the class considered here.

The limitation is therefore informational rather than computational. Portfolio directions deep in the managed-payoff spectrum are only weakly identified by a finite history of realized returns. A richer model may represent more investment opportunities, and a better algorithm may use the available information more effectively, but neither can make those weakly identified opportunities more visible in the data.

Learning faster requires knowing more about the economy. This can come from a longer market history or from additional economic restrictions on the return process, the managed-payoff spectrum, or the location of optimal portfolio demand. Absent such information, the learning rate is a

property of the investment problem, not of the particular algorithm used to solve it.

V. Representation, Learnability, and the Virtue of Complexity

Kelly et al. (2024) study an environment in which expanding the model also captures an increasing share of the underlying economic signal. In their setting, the improvement in approximation can dominate the additional estimation cost. This is the virtue of complexity.

Our question is different. We hold the investor's information set fixed and vary only the richness of the portfolio representation. We take the view that a portfolio manager should make available all information she considers economically useful and can observe in real time. Regularization, rather than ex ante feature exclusion, should then determine how much of the resulting portfolio opportunity set can actually be learned from the available data. We therefore study the virtue of Complexity, as the virtue of richer representation, separately from the addition of new signals.

Our point in this section is that a richer portfolio representation and a better learned portfolio are two different objects.

Let $\mathrm{SR}^\star$ denote the maximum population Sharpe ratio over the space of L^2 portfolio policies,

$$\mathrm{SR}^\star := \sup_{W \in L^2} \mathrm{SR}(W),$$

and let

$$\mathrm{SR}_K^\star := \sup_{W \in \mathcal{H}_K} \mathrm{SR}(W)$$

denote the maximum population Sharpe ratio available within representation K, where $\mathcal{H}_K \subseteq L^2$ is the corresponding policy space. For example, a linear kernel restricts portfolio weights to be linear functions of characteristics, whereas a Matérn kernel allows a richer nonlinear class.

Finally,

$$\mathrm{SR}(\widehat{W}_{K,\lambda,T})$$

denotes the population Sharpe ratio of the portfolio policy learned within representation K from a market history of length T.

The gap between the unrestricted population opportunity set and the portfolio actually learned from the data can be decomposed as

$$\mathrm{SR}^\star - \mathrm{SR}(\widehat{W}_{K,\lambda,T}) = \underbrace{\mathrm{SR}^\star - \mathrm{SR}^\star_K}_{\text{representation gap}} + \underbrace{\mathrm{SR}^\star_K - \mathrm{SR}(\widehat{W}_{K,\lambda,T})}_{\text{learning gap}}. \tag{12}$$

The representation gap measures the investment opportunities that cannot be expressed by the chosen representation, even if the investment environment were known perfectly. The learning gap measures how much of the opportunity set available within that representation cannot be recovered from a finite history of returns.

Now consider two nested representations,

$$\mathcal{H}_{K_1} \subseteq \mathcal{H}_{K_2}.$$

At the population level,

$$\mathrm{SR}^\star_{K_2} \geq \mathrm{SR}^\star_{K_1}.$$

A richer representation can therefore only expand the population investment opportunity set. In this precise sense, complexity is a virtue for representation.

But this population ordering need not survive estimation from finite data. It is perfectly possible that

$$\mathrm{SR}^\star_{K_2} > \mathrm{SR}^\star_{K_1}$$

while

$$\mathrm{SR}(\widehat{W}_{K_2,\lambda_2,T}) < \mathrm{SR}(\widehat{W}_{K_1,\lambda_1,T}).$$

The larger representation contains better population opportunities, but some of those opportunities may lie in managed-portfolio directions that are too weakly identified by the available return history.

A richer representation therefore expands what the portfolio manager could exploit. Whether

those additional opportunities improve the learned portfolio depends on whether they can actually be learned from the available data.

VI. Empirical Portfolio Learnability

A. Data, Representation, and Investment Performance

The empirical exercise asks how a representation of stock characteristics, finite return history, and regularization jointly shape a portfolio policy. A richer representation makes additional portfolio opportunities available, but a finite return history does not necessarily make all those opportunities learnable or financially valuable. We first compare Linear, Gaussian, and Matérn-$3/2$ policies on identical subsequent months. We then fix the Gaussian representation to examine regularization and the economic content of its additional directions. The evidence is descriptive historical evidence: neither a favorable Sharpe nor an estimated number of active directions certifies a reliably profitable opportunity.

We use the monthly U.S. equity characteristic panel of Jensen, Kelly, and Pedersen, with formations from January 1963 through December 2024. Starting from their research dictionary, we exclude nano stocks, retain the 130 characteristics with greatest coverage in 1963–1972, and remove observations missing more than 30% of these inputs. Observed values receive average cross-sectional ranks on $[-0.5, 0.5]$; remaining missing inputs receive zero. The panel is unbalanced. The retrospectively constructed characteristic dictionary and current data snapshot constitute the academic information set, rather than a reconstruction of historical signal discovery or data vintages.

There is a material sample limitation. Unavailable subsequent payoffs are excluded before ranking and calculating the number of stocks N_t. A formation-only audit leaves 7,489 unidentified stock-month payoffs out of 2,504,402 eligible observations (0.299%); documented adjacent-calendar source matches recover none. Formation-only ranks are invariant to perturbing future payoff availability, whereas the retained complete-payoff sample is conditional on that availability. Thus selection uses only earlier observed payoff values, but sample formation still depends on whether future payoffs are available. Unknown returns are never assigned zero. All performance

and cost results retain this caveat.

The first inner training period uses 1963–1972 formations and validation uses 1973–1977. Each subsequent January-close decision reserves the preceding 60 formation months for validation, then refits on the full available pretest history. December's payoff is known at January close. January–December formation positions therefore realize February–next January returns. All three policies are evaluated on the same 564 OOS months, February 1978–January 2025.

Linear uses a constant and the characteristic coordinates. Gaussian and Matérn-$3/2$ use 10,000 random Fourier features with seed 0. For each kernel, chronological validation jointly selects lengthscale and ridge penalty by minimizing the mean response-one quadratic loss Q at fitted scale. Lengthscales are $\{0.25, 0.5, 1, 2, 4\}$ times the initial-sample median distance; each has 120 penalties fixed from its own initial training spectrum. Frequency and phase draws are shared across lengthscales. The chosen pair is refitted using training and validation observations before the test year; no test payoff selects either parameter. Linear selects its penalty by the same chronological loss criterion. This is a comparison of representations under past validation; finite feature banks do not establish strict nesting or identify a population approximation gap from an observed performance ranking.

Table I. Representation comparison under chronological validation

Policy	Mean %	Vol. %	SR_G	SR_N	Gross	Trades	Fees %
Linear	23.44	7.58	3.091	2.476	2.01	19.45	5.14
Gaussian	28.34	7.77	3.646	2.857	2.44	24.53	6.48
Matérn-$3/2$	28.85	8.04	3.587	2.747	2.58	26.82	7.07

All policies use the same 564 months, February 1978–January 2025, and common initial-validation exposure scale. Gaussian and Matérn jointly validate lengthscale and penalty; Linear validates penalty. Mean and volatility are annualized gross excess returns. Net Sharpe applies 25 bps two-sided commissions and 30 bps/year short fees. Gross is mean account gross/NAV; trades are annual drift-adjusted traded notional/NAV on the net path; fees are annual trading plus short charges/NAV. Costs are illustrative assumptions.

A common positive scale, 0.0374000, maps the initial Linear validation policy's median gross exposure to 1.8. It is fixed before the first test payoff and applies to all policies, without future volatility or full-sample exposure normalization. It leaves gross Sharpe unchanged but matters for

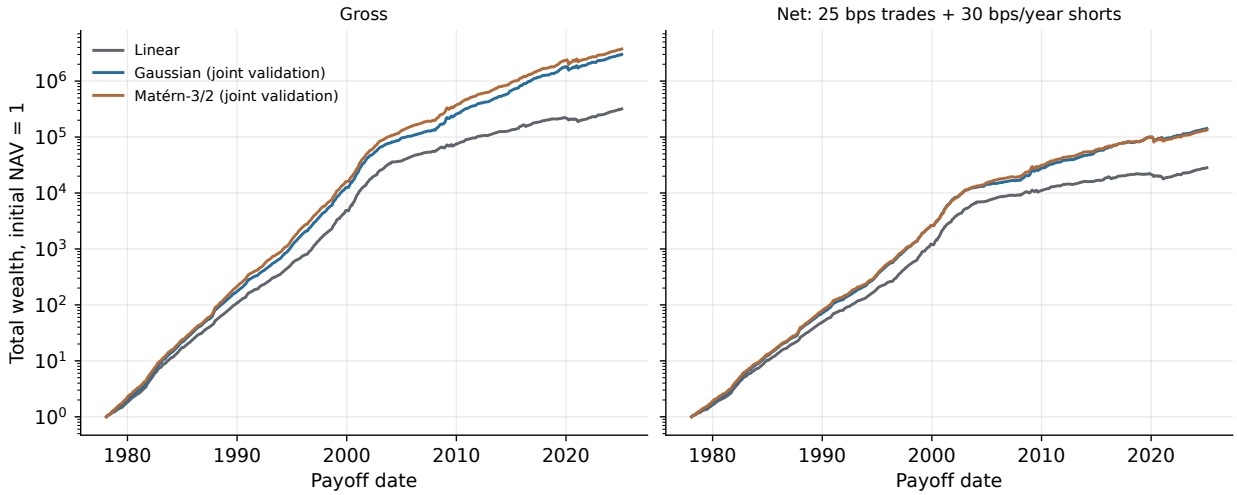


Figure 3. Investment performance on identical OOS months. Linear validates its ridge penalty; Gaussian and Matérn-3/2 jointly validate lengthscale and penalty using earlier observations. Total wealth compounds cash returns plus portfolio excess payoffs, starting from NAV one at January 1978 close. The right panel applies 25-basis-point commissions to both purchases and sales against drift-adjusted holdings, plus a 30-basis-point annual short fee. These are illustrative costs on a sample conditional on future-payoff availability.

costs. The Gaussian-minus-Linear gross Sharpe difference is 0.554, with paired 12-month-block interval [0.275,0.880]; the net difference is 0.381, with interval [0.140,0.653]. The corresponding Matérn differences are 0.496 [0.133,0.907] gross and 0.271 [-0.026,0.611] net. These comparisons condition on the fitted paths and use the same resampled months for each policy. Trading requirements and uncertainty accompany the point estimates; the comparison does not establish uniform dominance of one representation.

B. A Fixed Economic Spectrum and Active Complexity

For the remainder of the section we use the existing fixed-lengthscale Gaussian specification: initial median distance 4.392108, 10,000 seed-0 features, and the original past-validation ridge selections. Its full-policy gross/net Sharpes are 3.300/2.605. These are distinct from the jointly validated Gaussian returns in Figure 3 and Table I. Fixing lengthscale and feature draws keeps the opportunity representation unchanged: varying λ then changes regularization within the same managed-payoff

operator. A lengthscale change would alter both that operator and the portfolios being decomposed.

Let G contain historical monthly managed payoffs $X_t = N_t^{-1} \sum_i R^e_{i,t+1} \phi(Z_{i,t})$. Response-one ridge fits $a_\lambda = (G'G/T + \lambda I)^{-1} G'\mathbf{1}/T$ for positive λ. The uncentered second-moment spectrum of $G'G/T$ orders directions by historical payoff variation. An eigenvalue is neither invested capital, expected return, Sharpe, nor a uniquely named anomaly. Figure 4 uses only the history available at January 2024 close. Raw eigenvalues preserve representation units; division by the trace is a concentration diagnostic that does not rescale the fitted policy.

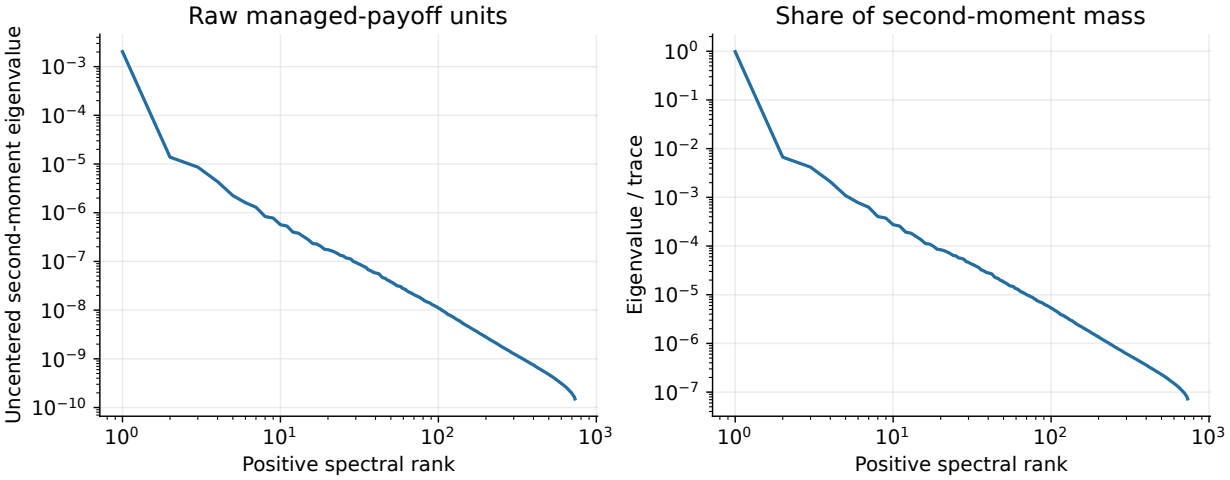


Figure 4. Managed-payoff spectrum of the fixed Gaussian representation at January 2024 close. Formations through December 2023 give 732 known monthly payoffs. The left panel shows raw uncentered second-moment eigenvalues; the right divides by their trace. Both use the same fixed initial median lengthscale and feature bank as Figures 5–7 and the spectral accounts in Table III.

Mass concentration and active complexity are distinct. Ridge weights a direction by $\widehat{\mu}_j/(\widehat{\mu}_j+\lambda)$, giving $C(\lambda) = \sum_j \widehat{\mu}_j/(\widehat{\mu}_j+\lambda)$. Weak eigenvalues can contribute many effective degrees of freedom while carrying little second-moment mass. Across annual histories, the first 10% of positive Gaussian ranks contain 99.921% of mass, but contribute only 33.8 of the selected policy's 124.5 mean effective directions. Positive rank uses a relative 10^{-12} eigenvalue tolerance; near-tied eigenvalues are kept together at group boundaries. Feasible complexity cannot exceed this historical effective rank, which is at most the number of historical monthly observations. A long weak tail can change

exposures without establishing that its opportunities are learnable.

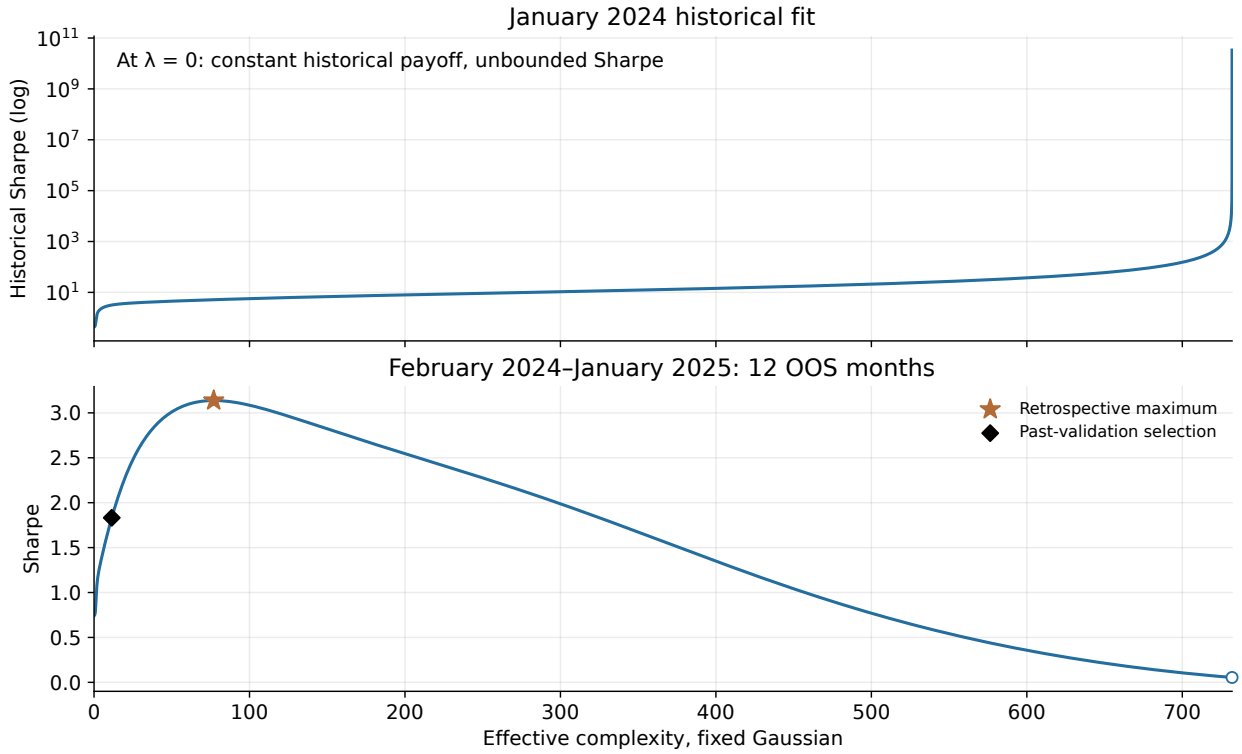


Figure 5. Complete regularization path for the fixed Gaussian January 2024 refit. Historical Sharpe uses a logarithmic scale; subsequent Sharpe uses a separate linear scale. Exact response-one interpolation has zero historical variance and unbounded historical Sharpe; unresolved numerical ratios near that limit are omitted from the historical panel only. Each positive-penalty point is evaluated on its actual February 2024–January 2025 OOS payoffs. The open circle is the exact minimum-norm zero-penalty policy at effective rank 732. The star is the retrospective maximum over the extended evaluation grid; the diamond retains the original earlier-validation selection. Neither is an estimate of a significant population optimum.

The final-year path holds lengthscale, feature bank, and history fixed. Its retrospective maximum is at complexity 77.0 with subsequent Sharpe 3.138; at zero penalty Sharpe is 0.054. Historical fit and subsequent investment performance need not move together. Twelve subsequent months produce a noisy illustration, so Figure 6 provides the complete progression in nine historical blocks.

The evaluation grid retains every original penalty and adds a logarithmic sequence from strong regularization to very small positive penalties, followed by exactly $\lambda = 0$. Its 1,522 policies are evaluated separately at every annual refit. The zero endpoint uses a symmetric eigensystem Moore–

Penrose pseudoinverse with the same positive-rank tolerance, rather than an inverse of a singular operator. Direct feature-space least squares checks the first and final endpoints, and the smallest positive penalties converge numerically to the zero returns. No added evaluation changes the original validation policy, and no OOS result selects a deployable penalty.

To assess whether the final-year Sharpe–complexity profile is representative of the broader historical record, Figure 6 repeats the analysis across nine nonoverlapping investment periods. Holding the Gaussian representation fixed, we vary only the degree of regularization and evaluate each candidate on subsequent portfolio returns. The profiles show that realized performance need not improve monotonically as additional portfolio directions are activated. In several periods, Sharpe reaches an interior maximum and subsequently declines, although the magnitude of the decline varies considerably across investment histories.

The annual cross-validation portfolio remains an implementable, past-selected policy, but it is not a single point on the block-level fixed-penalty path because the selected penalty can change from one annual refit to the next.

Table II. Retrospective maxima and exact unregularized endpoints

Decision years	C at max.	Max. SR	$C(0)$	SR at $\lambda = 0$	Decline
1978–1982	113.4	5.719	204	5.313	0.406
1983–1987	209.1	6.219	264	6.134	0.085
1988–1992	137.9	7.626	324	6.274	1.352
1993–1997	158.2	7.608	384	6.609	0.999
1998–2002	164.4	5.443	444	4.844	0.599
2003–2007	165.4	2.970	504	2.410	0.559
2008–2012	28.7	2.918	564	2.435	0.483
2013–2017	379.2	3.228	624	3.158	0.070
2018–2024	266.2	1.537	696	1.337	0.200

All maxima use the entire extended grid. Complexity is the realized-month-weighted mean across annual refits; $C(0)$ equals mean historical effective rank. Decline is maximum minus zero-penalty Sharpe, a descriptive change rather than a significance test. The smallest-decline period is retained as observed.

All nine retrospective maxima are interior on the extended grid. Maximizing mean complexity ranges from 28.7 to 379.2. The 2008–2012 maximum occurs at low complexity. The unregularized

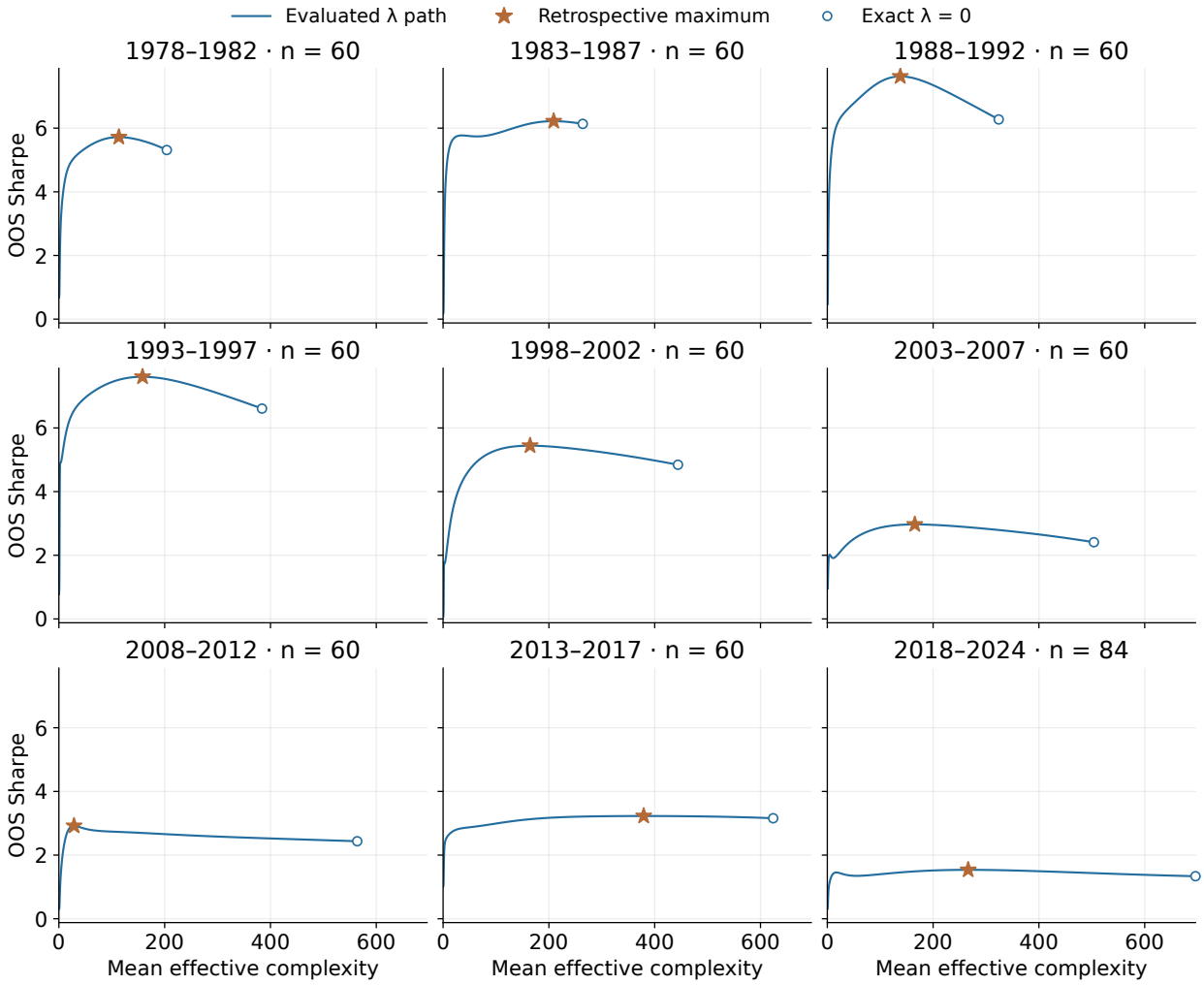


Figure 6. Historical out-of-sample Sharpe–complexity profiles. Each panel reports the realized out-of-sample Sharpe ratio associated with the complete ridge regularization path of the fixed-lengthscale Gaussian representation. The nine panels cover eight five-year decision blocks and one 2018–2024 block. For each candidate penalty, monthly out-of-sample returns are concatenated across the annual refits in the block, and a single Sharpe ratio is computed. Effective complexity is averaged across those refits, weighted by the number of evaluated months. Curves connect the actual evaluated penalties, including the extended positive grid and the exact minimum-norm zero-penalty endpoint, without smoothing or extrapolation. Stars indicate retrospective test-sample maxima; open circles indicate the unregularized endpoints. The dynamically selected annual cross-validation strategy is not plotted as a point on these curves because its penalty may differ across annual refits. All panels use common axes. The historical maxima are descriptive and are not estimates of population-optimal complexity.

endpoints remain close to the maxima in 1983–1987 and 2013–2017, with peak-to-zero Sharpe declines of 0.085 and 0.070, respectively. Neither period displays a substantial endpoint deterioration. The 2018–2024 decline is also modest; we do not impose a common downturn. Across blocks, the endpoint decline ranges from 0.070 to 1.352.These are descriptive shapes, not evidence of a statistically identified optimum in every period. Broad regions can be nearly flat relative to sampling uncertainty. All displayed maxima have been recomputed on the extended grid, including zero.

Intervals use 5,000 paired circular-block resamples, sharing monthly draws across all candidates or contrasted portfolios. They condition on estimated policies, representations, and holdings; they do not repeat model selection. Approximate stability and dependence decay within a block motivate this diagnostic. It neither corrects for searching the grid nor establishes validity under arbitrary nonstationarity. Six- and 24-month-block results for the representation comparisons and economic contrasts accompany the canonical results. A favorable retrospective peak is consequently distinct from a learnable policy selected using past information.

C. What Additional Directions Trade

We interpret portfolios associated with eigenspaces of the same fixed Gaussian policy. At each annual refit, four disjoint positive-rank groups partition its ridge coefficients. Their actual contributions are reconstructed as stock-level signed holdings, providing long and short characteristic profiles and subsequent payoff and covariance attribution. The same selected penalty is retained across nested comparisons. These are attribution portfolios within one policy, not independently optimized strategies. Near-tied eigenvectors can rotate, so group profiles carry the historical interpretation rather than persistent names assigned to individual eigenvectors.

Leading holdings favor profitability and oppose recent returns and debt financing. Intermediate directions modify those exposures within the same characteristic universe. A large normalized characteristic contrast can coexist with small notional exposure, so profiles and signed notionals answer different questions. Formation descriptors characterize what is held; subsequent outcomes

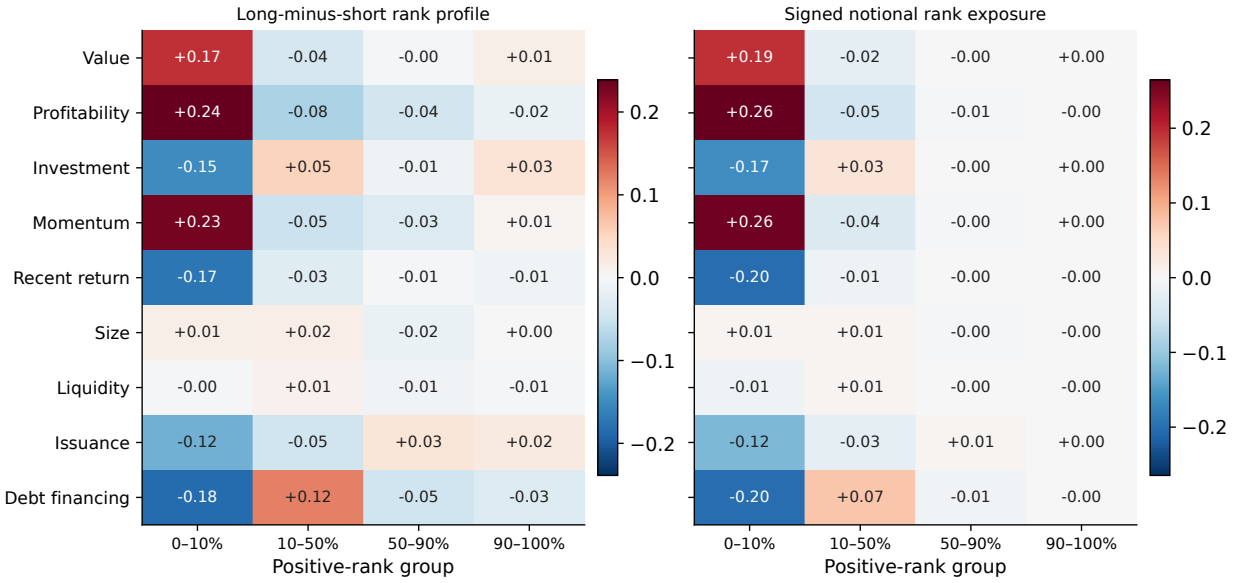


Figure 7. Economic profiles of the fixed Gaussian policy's actual spectral holdings. The left panel gives long-minus-short weighted formation-rank scores; the right preserves investment scale through signed notional exposures. An explicit signed characteristic dictionary defines the families. Debt financing and equity issuance are distinct. Families can overlap and cannot be added as disjoint return components. These holdings decompose the fixed representation, independently of the tuned-kernel opening comparison.

determine its investment contribution.

Table III. Fixed Gaussian spectral groups and incremental investment value

Ranks	Mass %	Own gross	Account gross	Mean %	SR_G	SR_N
0–10%	99.9215	2.267	2.267	24.323	3.012	2.354
10–50%	0.0685	1.263	2.333	2.759	3.284	2.598
50–90%	0.0091	0.327	2.325	0.047	3.286	2.594
90–100%	0.0010	0.078	2.326	0.026	3.300	2.605

Added ranks	ΔQ	Cross term	Own square	Paired 95% interval
0–10%	-0.402340	-1.083912	0.681573	[-0.524808, -0.257425]
10–50%	-0.036008	-0.131766	0.095758	[-0.093264, +0.027912]
50–90%	+0.000168	-0.003978	0.004146	[-0.008734, +0.009000]
90–100%	-0.003123	-0.003547	0.000424	[-0.006796, -0.000185]

The original selected penalty is retained at each annual refit. Mass averages annual historical second-moment shares. Own gross and mean describe individual components; account gross and Sharpes describe all groups through that boundary. The lower panel reports raw fitted-scale loss changes, relative to the null portfolio for the first group, with the same 5,000 paired 12-month-block draws. Weights are summed at stock level before trades and short fees. Gross Sharpe increments are +0.272, +0.002, +0.014; net increments are +0.244, -0.004, +0.011.

The leading group has mean component gross 2.267 and annual excess payoff 24.32%. Adding intermediate directions contributes 2.76% annually. Their own gross 1.263 increases the cumulative account's gross by only 0.066: some added positions offset existing stock holdings. Component gross exposures therefore do not add to the combined account's gross. The component's annual covariance with the leading portfolio is -0.000634, while total annual variance changes by 0.000278. Negative covariance can coexist with increased account variance because the component also adds its own variance.

Gross Sharpe rises from 3.012 to 3.284, and net Sharpe from 2.354 to 2.598. Paired 12-month-block intervals for these changes are [0.002,0.546] and [0.011,0.487]. The response-one loss changes by -0.036008, with interval [-0.093264,0.027912]; its cross term is -0.131766 and own-square term 0.095758. Thus a positive standalone mean, a Sharpe increment, and an incremental Q improvement are distinct assessments. The incremental identity $\Delta Q = -2\,\overline{(1-P)A} + \overline{A^2}$ includes the interaction of the previous policy P and added payoff A, at their original fitted scale.

Small incremental account gross does not imply small turnover. Monthly characteristics and annual refits can change offsetting positions substantially. For every nested account, stock weights are summed before calculating trades against drift-adjusted holdings, short notional, and net returns. Costs begin with entry from cash. Target weights refer to post-execution NAV; proportional commissions are solved jointly with the target cash residual. The annual short fee is divided by twelve. Cash earns or pays the same monthly risk-free rate, and terminal holdings are marked rather than forcibly liquidated. Separate component fees would not reconstruct the netted account. Changes in target weights remain a separate turnover proxy, never observed execution costs.

Corporate-action and delisting treatment relies on the upstream JKP return construction and stable identifiers; there is no independent share-count or execution-tape reconstruction. Liquidated names incur trades but receive no invented next payoff. The cost assumptions omit market impact and additional funding spreads. They are self-financing friction sensitivities, with the same future-payoff-availability caveat as gross returns.

The weak tail has little historical second-moment mass and small absolute payoff contributions. The successive gross/net Sharpe increments in Table III include adverse as well as favorable outcomes; the loss decomposition also retains the own-square and cross terms. Small mass alone cannot rule out covariance value, while a striking normalized characteristic tilt need not have a material financial contribution. Six-benchmark regressions of the actual component returns give descriptive spanning statistics; a separate hedge uses only pretest slope estimates. Neither supplies an independently selected investment strategy or a newly identified risk premium.

Economic content and supported complexity vary with history. The stationary theory is a benchmark for approximately stable conditions; a local interpretation requires those conditions. Descriptive complexity changes do not identify a population spectral-decay parameter. Across the representation comparison, fixed-kernel regularization paths, and stock-level attribution, the financial question is whether additional available directions improve the portfolio that finite historical information can support.

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A. Proofs and Auxiliary Results

A.1. Maximum-Sharpe direction

Proof of Proposition 1. Write $W = aU$, with $\|U\|_{\mathcal{H}} = 1$ and $a \in \mathbb{R}$. For a direction with positive payoff variance, put $m_U = \mathbb{E}[R^p_{t+1}(U)]$ and $s^2_U = \mathbb{E}[(R^p_{t+1}(U))^2]$. Then

$$Q(aU) = 1 - 2am_U + a^2 s^2_U, \qquad a^\star(U) = \frac{m_U}{s^2_U}, \qquad \min_a Q(aU) = \frac{1}{1 + \mathrm{SR}^2(R^p_{t+1}(U))}.$$

A zero payoff has loss one; a nonzero constant payoff can be scaled to have loss zero. The assumption $0 < Q(W^\star) < 1$ therefore excludes both from the optimum. Minimizing over directions maximizes squared Sharpe, and the optimally scaled payoff has mean $m^2_U/s^2_U > 0$. Thus the selected orientation attains the positive maximum Sharpe ratio. □

A.2. Pricing-kernel interpretation and Hansen–Jagannathan equality

Proof of Proposition 2. Suppress the economy subscript. The first-order condition for $Q(W^\star + aV)$ at $a = 0$ gives $\mathbb{E}[m^\star R^p(V)] = 0$ for every $V \in \mathcal{H}$. Taking $V = W^\star$ and using Proposition 1,

$$\mathbb{E}[m^\star] = \mathbb{E}[(m^\star)^2] = Q(W^\star) =: q^\star = \frac{1}{1 + (\mathrm{SR}^\star)^2}.$$

Hence $\widetilde{m} = m^\star/q^\star$ has mean one and variance $(1 - q^\star)/q^\star = (\mathrm{SR}^\star)^2$. For any square-integrable d with $\mathbb{E}[d] = 1$ and $\mathbb{E}[dR^p(V)] = 0$ for every $V \in \mathcal{H}$, $\mathbb{E}[dm^\star] = 1$ and therefore

$$\mathrm{Var}(d) = \mathrm{Var}(\widetilde{m}) + \mathbb{E}[(d - \widetilde{m})^2] \geq (\mathrm{SR}^\star)^2.$$

This proves the minimum-variance property without imposing positivity or the stronger conditional restriction in Condition (E4). □

A.3. Managed-portfolio Markowitz representation

Proof of Proposition 3. The reproducing property gives $R^p_{t+1}(W) = \langle W, X_t\rangle_{\mathcal{H}_K}$ and $\|X_t\|^2_{\mathcal{H}_K} = N^{-2}R'_{t+1}K_tR_{t+1}$. Assumption 1 gives strong measurability, while Condition (E2) and Jensen's inequality give $\mathbb{E}\|X_t\|^2_{\mathcal{H}_K} \le B_X^2 \log 2$. Thus the first and second moments are well defined, and

$$Q(W) = 1 - 2\langle W, \mu_{\mathcal{H}_K}\rangle_{\mathcal{H}_K} + \langle W, \Sigma_{\mathcal{H}_K} W\rangle_{\mathcal{H}_K}.$$

The population first-order condition is $\Sigma_{\mathcal{H}_K} W^\star_K = \mu_{\mathcal{H}_K}$; the minimum-norm solution is the stated pseudoinverse representation. The empirical penalized criterion gives

$$(\widehat{\Sigma}_T + \lambda I)\widehat{W}_{\lambda,T} = \widehat{\mu}_T.$$

Since $\widehat{\Sigma}_T \ge 0$ and $\lambda > 0$, the inverse exists and yields (5). □

A.4. A sufficient Gaussian return specification

Proposition 4 (A sufficient Gaussian return specification)**.** *Suppose that, conditionally on $\mathcal{F}_t$, R_{t+1} is Gaussian, possibly with a singular covariance, and*

$$\max_{1\le i\le N} \mathbb{E}[R^2_{i,t+1} \mid \mathcal{F}_t] \le V_R < \infty \quad a.s.,$$

uniformly in the economy and time. Then Condition (E2) *holds with $B_X^2 = 8\kappa_K V_R$ and $K_4 = 3$.*

Proof. For every $V \in \mathcal{H}_K$, $R^p_{t+1}(V)$ is conditionally Gaussian. A scalar $G \sim N(m, v)$ satisfies

$$\mathbb{E}G^4 = m^4 + 6m^2v + 3v^2 \le 3(m^2 + v)^2,$$

which proves (8) with $K_4 = 3$.

For the tail condition, Cauchy–Schwarz in $\mathcal{H}_K$ and $K(z,z) \le \kappa_K$ give

$$\|X_t\|_{\mathcal{H}_K}^2 \le \frac{\kappa_K}{N} \sum_{i=1}^{N} R_{i,t+1}^2.$$

If $G \sim N(m, v)$ and $m^2 + v \le V_R$, then

$$\mathbb{E}e^{G^2/(8V_R)} = \left(1 - \frac{v}{4V_R}\right)^{-1/2} \exp\left(\frac{m^2}{8V_R - 2v}\right) \le \frac{2}{\sqrt{3}} e^{1/6} < 2.$$

Therefore, by convexity,

$$\begin{aligned}\mathbb{E}_\varepsilon\left[\exp\left(\frac{\|X_t\|^2}{8\kappa_K V_R}\right)\middle|\mathscr{F}_t\right] &\le \mathbb{E}_\varepsilon\left[\exp\left(\frac{1}{N}\sum_{i=1}^{N}\frac{R_{i,t+1}^2}{8V_R}\right)\middle|\mathscr{F}_t\right] \\ &\le \frac{1}{N}\sum_{i=1}^{N}\mathbb{E}_\varepsilon\left[\exp\left(\frac{R_{i,t+1}^2}{8V_R}\right)\middle|\mathscr{F}_t\right] < 2.\end{aligned}$$

Independence across stocks is unnecessary. □

This is a sufficient specification for Condition (E2); Gaussianity is not imposed on the class $\mathcal{E}$.

A.5. Exact reduction to the kernel learning problem

Lemma 1 (Induced KRR problem)**.** *For every $\varepsilon \in \mathcal{E}$, the direct estimator is isometrically equivalent to scalar KRR with constant response one. The induced problem satisfies the conditions of Bignozzi et al. (2026, Lemma A.7) with source order $r = 1$. Its covariance operator is unitarily equivalent to $\Sigma_{\mathcal{H}_K,\varepsilon}$, and its prediction error is exactly population portfolio regret.*

Proof. Step 1: identify the learning problem. Let $\mathcal{X} = \mathcal{Z}^N \times \mathbb{R}^N$ and $\Xi_t = (Z_t, R_{t+1})$. For $x = ((z_1, \dots, z_N), r)$, put

$$\Phi(x) = \frac{1}{N}\sum_i r_i K(\cdot, z_i), \qquad f_W(x) = \langle W, \Phi(x)\rangle_{\mathcal{H}_K}.$$

The map $\mathcal{T} : W \mapsto f_W$ is injective: choosing $r = (N, 0, \ldots, 0)$ and varying z_1 shows that $f_W = 0$ on $\mathcal{X}$ implies $W = 0$. Equip $\mathcal{G} = \{f_W : W \in \mathcal{H}_K\}$ with $\langle f_W, f_V \rangle_{\mathcal{G}} = \langle W, V \rangle_{\mathcal{H}_K}$. Then $\mathcal{T}$ is an isometry onto the separable RKHS with kernel

$$\widetilde{K}(x, x') = \langle \Phi(x), \Phi(x') \rangle_{\mathcal{H}_K} = \frac{1}{N^2} \sum_{i,j} r_i s_j K(z_i, z'_j), \qquad x' = ((z'_1, \ldots, z'_N), s).$$

Since $\Phi(\Xi_t) = X_t$, $Y_t \equiv 1$ gives exactly the empirical criterion in (3), with the same penalty. Let $J : \mathcal{G} \to L^2(\mathbb{P}_{\varepsilon,\Xi})$ be the canonical embedding. Square integrability, established above, gives

$$\mathcal{T}^* J^* J \mathcal{T} = \Sigma_{\mathcal{H}_K, \varepsilon}, \qquad f_{\mathcal{G}} = \mathcal{T} W^\star_\varepsilon.$$

Expansion of the population criterion at its minimizer gives

$$\|J f_W - J f_{\mathcal{G}}\|^2_{L^2} = \|\Sigma^{1/2}_{\mathcal{H}_K, \varepsilon}(W - W^\star_\varepsilon)\|^2_{\mathcal{H}_K} = Q_\varepsilon(W) - Q_\varepsilon(W^\star_\varepsilon).$$

Step 2: verify the statistical assumptions. The induced kernel is measurable, and $\widetilde{K}(\Xi_t, \Xi_t) = \|X_t\|^2_{\mathcal{H}_K}$. For the statistical history $\mathcal{A}_t = \sigma(\Xi_s, Y_s : s \le t)$, economic timing gives $\mathcal{A}_{t-1} \subseteq \mathcal{F}_t$. Condition (E2) and the tower property therefore imply

$$\mathbb{E}_\varepsilon \left[\exp\left(\frac{\widetilde{K}(\Xi_t, \Xi_t)}{B_X^2} \right) \middle| \mathcal{A}_{t-1} \right] \le 2.$$

This is the conditional kernel-moment assumption in Bignozzi et al. (2026), with $B_K = B_X$ and $C_K = 2$. Stationarity and mixing follow from Condition (E1); Condition (E3) gives source order $r = 1$, and Condition (E5) gives the same positive spectrum under the isometry.

It remains to verify the score condition of the cited lemma. Write $m^\star = m^\star_{t+1,\varepsilon}$, $P^\star_t = R^p_{t+1}(W^\star_\varepsilon)$ and $q_t = \mathbb{E}_\varepsilon[(P^\star_t)^2 \mid \mathcal{F}_t]$. Condition (E4) with $V = W^\star_\varepsilon$ gives $\mathbb{E}_\varepsilon[P^\star_t \mid \mathcal{F}_t] = q_t$; conditional Jensen then gives $q_t^2 \le q_t$, so $0 \le q_t \le 1$. Consequently, Condition (E2) implies $\mathbb{E}_\varepsilon[(m^\star)^4 \mid \mathcal{F}_t] \le$

$8(1 + K_4)$. For every $V \in \mathcal{H}_K$, conditional Cauchy–Schwarz yields

$$\begin{aligned}\mathbb{E}_\varepsilon[(m^\star)^2(R^p_{t+1}(V))^2 \mid \mathscr{F}_t] &\le \sqrt{\mathbb{E}_\varepsilon[(m^\star)^4 \mid \mathscr{F}_t]\mathbb{E}_\varepsilon[(R^p_{t+1}(V))^4 \mid \mathscr{F}_t]} \\ &\le C_{\mathrm{sc}}\mathbb{E}_\varepsilon[(R^p_{t+1}(V))^2 \mid \mathscr{F}_t],\end{aligned}$$

where $C_{\mathrm{sc}} = \sqrt{8K_4(1 + K_4)}$. Integration and summation over an orthonormal basis show that $m^\star X_t$ is square integrable and that its second-moment operator is bounded above by $C_{\mathrm{sc}}\Sigma_{\mathcal{H}_K,\varepsilon}$. Condition (E4), separability and the tower property give $\mathbb{E}_\varepsilon[m^\star X_t \mid \mathcal{A}_{t-1}] = 0$. The statistical score is $\mathcal{T}(m^\star X_t)$, so the cited lemma applies with its variance constant equal to C_{sc}. All constants are common to $\mathcal{E}$. □

A.6. Proof of the portfolio-learning rate

Proof of Theorem 1. Lemma 1 verifies the conditions of Bignozzi et al. (2026, Lemma A.7) with $r = 1$. Since $\gamma > b$ and $\lambda^\star_T \asymp T^{-b/(b+1)}$, its upper rate and the loss identity give (9) uniformly over $\mathcal{E}$.

For the Sharpe statement, put $D_T = Q_\varepsilon(\widehat{W}_{\lambda^\star_T,T}) - Q_\varepsilon(W^\star_\varepsilon)$, $q^\star = Q_\varepsilon(W^\star_\varepsilon)$ and $S^\star = \mathrm{SR}^\star_{\varepsilon,K}$. Proposition 1 and Condition (E6) give

$$\underline{q} := \frac{1}{1 + \overline{S}^2} \le q^\star \le \frac{1}{1 + \underline{S}^2} =: \overline{q} < 1.$$

On $D_T < 1 - \overline{q}$, the estimated loss is below one, hence the estimated payoff has positive mean. Its variance cannot be zero: a positive constant payoff could be rescaled to one, contradicting $q^\star > 0$. Write its positive Sharpe as $\widehat{S}_T$. Optimizing only its scale gives

$$q^\star \le \frac{1}{1 + \widehat{S}^2_T} \le q^\star + D_T, \qquad 0 \le (S^\star)^2 - \widehat{S}^2_T \le \frac{D_T}{q^\star(q^\star + D_T)}.$$

Therefore

$$0 \le S^\star - \widehat{S}_T \le \frac{D_T}{\underline{S}\,\underline{q}^2}.$$

The event has probability tending to one uniformly, which proves (10). Any finite convention on a zero-variance event leaves this rate unchanged. At a fixed volatility σ, the capital-allocation-line gap is $\sigma(S^\star - \widehat{S}_T)$ and has the same rate, uniformly for σ in a bounded interval. □

A.7. Effective portfolio complexity

Proof of Theorem 2. Condition (E5) and Bignozzi et al. (2026, Proposition A.6) give $C_{K,\varepsilon}(\lambda) \asymp \lambda^{-1/b}$ uniformly. At the regularization scale in Theorem 1,

$$C^\star_{K,T} \asymp T^{1/(b+1)}, \qquad \frac{C^\star_{K,T}}{T} \asymp T^{-b/(b+1)} \longrightarrow 0.$$

Finiteness follows from $C_{K,\varepsilon}(\lambda) \le \mathrm{Tr}(\Sigma_{\mathcal{H}_K,\varepsilon})/\lambda$. This is the rate-optimal balance of the loss upper bound, not a shape theorem for the exact Sharpe curve of every economy. □

A.8. A finance-specific minimax construction

Proof of Theorem 3. The construction keeps the policy kernel fixed. It is a least-favourable family used only in this lower-bound proof, not a structural return equation imposed on the economies in the main text.

Step 1: construct a least-favourable family with nondegenerate Sharpe. Since $\mathcal{E}$ is nonempty, take the one-period state-return marginal ν_0 of one economy in the class. Write a draw as $x = (\mathbf{z}, r)$ and put

$$Q_X(x) := \|\Phi(x)\|^2_{\mathcal{H}_K}, \qquad q_X := \int Q_X(x)\, d\nu_0(x).$$

Condition (E2) and Jensen's inequality imply $q_X \le B_X^2 \log 2$. Moreover $q_X > 0$, because otherwise the managed feature would vanish almost surely, contradicting the spectral lower bound in

Condition (E5).

Define a tilted law by

$$d\widetilde{\nu}(x) = \frac{Q_X(x)}{q_X}\, d\nu_0(x),$$

and, on the support of the tilted law, replace the return vector by

$$r^\circ = \sqrt{\frac{q_X}{Q_X(x)}}\, r.$$

Let ν denote the law of $(\mathbf{z}, r^\circ)$ and reuse r for the normalized return vector below. Since the feature map is linear in returns,

$$\int \Phi(\mathbf{z}, r^\circ) \otimes \Phi(\mathbf{z}, r^\circ)\, d\widetilde{\nu} = \int \Phi(x) \otimes \Phi(x)\, d\nu_0(x) =: \Sigma_0.$$

Thus the normalization preserves the entire managed-payoff second-moment operator. Under ν,

$$\|\Phi(x)\|^2_{\mathcal{H}_K} = q_X \le B_X^2 \log 2 \quad \text{almost surely.}$$

Put $\overline{B}_X := B_X\sqrt{\log 2}$.

Write

$$\Sigma_0 e_j = \mu_j e_j, \qquad c_\mu j^{-b} \le \mu_j \le C_\mu j^{-b}.$$

Its eigenvectors are orthonormal in $\mathcal{H}_K$. Fix constants $a, h > 0$, with $h \le a$, sufficiently small that

$$a^2 + h^2 \le R^2, \qquad \overline{B}_X\sqrt{a^2+h^2} \le \frac{1}{2}, \qquad 8C_\mu h^2 \le \frac{1}{8}.$$

For an integer $M \ge 1$ and $\theta \in \{-1, 1\}^M$, define

$$W_\theta := ae_1 + \frac{h}{\sqrt{M}} \sum_{j=M+1}^{2M} \theta_{j-M} e_j, \qquad g_\theta(x) := \langle W_\theta, \Phi(x)\rangle_{\mathcal{H}_K}.$$

Then

$$\|W_\theta\|_{\mathcal{H}_K} = \sqrt{a^2 + h^2} \le R, \qquad |g_\theta(x)| \le \overline{B}_X \sqrt{a^2 + h^2} \le \frac{1}{2}.$$

Put

$$A_M := \|g_\theta\|^2_{L^2(\nu)} = a^2 \mu_1 + \frac{h^2}{M} \sum_{j=M+1}^{2M} \mu_j.$$

This quantity does not depend on θ. Moreover,

$$a^2 c_\mu \le A_M \le \frac{1}{4}.$$

Step 2: construct admissible financial observations. Draw $x_t = (Z_t, r_t)$ independently with law ν. Conditional on x_t, draw $s_t \in \{-1, 1\}$ according to

$$\mathbb{P}_\theta(s_t = 1 \mid x_t) = \frac{1 + g_\theta(x_t)}{2}.$$

The observed financial row is $(Z_t, R_{t+1}) = (Z_t, s_t r_t)$. For the least-favourable construction take

$$\mathscr{F}_t = \sigma(x_u, s_u : u < t) \vee \sigma(x_t).$$

This information structure is allowed by Definition 1. Rows are i.i.d. The managed feature is $s_t \Phi(x_t)$ and, since $s_t^2 = 1$,

$$\|s_t \Phi(x_t)\|^2_{\mathcal{H}_K} = q_X.$$

Because x_t is contained in $\mathscr{F}_t$,

$$\mathbb{E}_\theta\left[\exp\left(\frac{\|s_t \Phi(x_t)\|^2_{\mathcal{H}_K}}{B_X^2} \right) \middle| \mathscr{F}_t \right] = \exp(q_X / B_X^2) \le 2.$$

Moreover, for every $V \in \mathcal{H}_K$,

$$(R^p_{t+1}(V))^2 = \langle V, \Phi(x_t)\rangle^2_{\mathcal{H}_K}, \qquad (R^p_{t+1}(V))^4 = \langle V, \Phi(x_t)\rangle^4_{\mathcal{H}_K}$$

conditionally on $\mathscr{F}_t$. Hence Condition (E2) holds with fourth-moment coefficient one, and therefore with the class constant $K_4 \geq 1$. The second-moment operator of the managed feature is exactly Σ_0.

For every $V \in \mathcal{H}_K$, conditional on the current x_t and the past,

$$\begin{aligned}&\mathbb{E}_\theta\big[(1 - R^p_{t+1}(W_\theta))R^p_{t+1}(V) \mid x_t, \mathscr{F}_{t-1}\big] \\ &\qquad = \langle V, \Phi(x_t)\rangle_{\mathcal{H}_K}\mathbb{E}_\theta[s_t - g_\theta(x_t) \mid x_t, \mathscr{F}_{t-1}] = 0.\end{aligned}$$

Thus Condition (E4) holds. For any portfolio policy W with finite loss, write $g_W(x) = N^{-1}\sum_i W(z_i)r_i$. Then

$$Q_\theta(W) - Q_\theta(W_\theta) = \int (g_W(x) - g_\theta(x))^2\, d\nu(x). \tag{A.1}$$

Indeed, conditioning on x gives the quadratic $1 - 2g_\theta(x)g_W(x) + g_W(x)^2$. Consequently, W_θ is the population-optimal minimum-norm policy.

Its payoff satisfies

$$\mathbb{E}_\theta[R^p_{t+1}(W_\theta)] = A_M, \qquad \mathbb{E}_\theta[(R^p_{t+1}(W_\theta))^2] = A_M.$$

Hence

$$(\mathrm{SR}^\star_\theta)^2 = \frac{A_M}{1 - A_M}.$$

Because A_M is uniformly bounded away from zero and from one, the optimal Sharpe ratio is uniformly bounded away from zero and infinity. Thus the least-favourable family satisfies Condition (E6), with constants common to the family, as well as Conditions (E1)–(E5).

Step 3: neighboring economies are difficult to distinguish. Consider the more informative experiment that also reveals x_t and s_t. A lower bound in this augmented experiment also applies to the

financial observations. Neighboring parameters differ in one sign, say coordinate $j - M$. Their conditional Bernoulli probabilities lie in $[1/4, 3/4]$, and

$$\mathrm{KL}(\mathrm{Ber}(p), \mathrm{Ber}(q)) \le \frac{(p-q)^2}{q(1-q)}.$$

Hence, for the laws $\mathbb{P}_\theta^{\mathrm{aug},T}$ of T augmented observations and a one-bit neighbor θ', we have

$$\begin{aligned}\mathrm{KL}(\mathbb{P}_\theta^{\mathrm{aug},T}, \mathbb{P}_{\theta'}^{\mathrm{aug},T}) &\le 2T \int (g_\theta - g_{\theta'})^2 \, d\nu \\ &= \frac{8Th^2\mu_j}{M} \\ &\le 8C_\mu h^2 \frac{T}{M^{b+1}}.\end{aligned}$$

Choose

$$M = \left\lceil T^{1/(b+1)} \right\rceil.$$

The right-hand side is at most $1/8$, so Pinsker's inequality gives total variation at most $1/4$ for every neighboring pair.

Step 4: testing difficulty implies scale-free directional error. Define

$$\psi_j(x) := \frac{\langle e_j, \Phi(x) \rangle_{\mathcal{H}_K}}{\sqrt{\mu_j}},$$

so that $(\psi_j)_{j \ge 1}$ is orthonormal in $L^2(\nu)$. Let $\widehat{g}$ be any estimated square-integrable payoff rule and write

$$\widehat{a}_j := \langle \widehat{g}, \psi_j \rangle_{L^2(\nu)}.$$

Decode the signs by

$$\widehat{\theta}_{j-M} := \mathrm{sign}(\widehat{a}_1 \widehat{a}_j), \qquad M < j \le 2M,$$

with any fixed convention at zero. This decoder is invariant to any nonzero global rescaling of $\widehat{g}$.

Define the scale-free directional error

$$D_\theta(\widehat{g}) := \inf_{c\in\mathbb{R}} \|g_\theta - c\widehat{g}\|^2_{L^2(\nu)}.$$

For any $c \in \mathbb{R}$, if $c\widehat{a}_1 \le 0$, the squared error in the first coordinate is at least $a^2\mu_1$. If $c\widehat{a}_1 > 0$, every incorrectly decoded sign contributes at least $h^2\mu_j/M$ to the squared error. Since $h \le a$ and $H(\widehat{\theta},\theta) \le M$, both cases imply

$$D_\theta(\widehat{g}) \ge \frac{h^2 c_\mu}{2^b M^{b+1}} H(\widehat{\theta},\theta).$$

For every neighboring pair, the sum of the two sign-error probabilities is at least $1 - \mathrm{TV} \ge 3/4$. Averaging over the hypercube and summing over its M coordinates therefore gives

$$2^{-M}\sum_\theta \mathbb{E}_\theta[H(\widehat{\theta},\theta)] \ge \frac{3M}{8}.$$

Consequently,

$$2^{-M}\sum_\theta \mathbb{E}_\theta[D_\theta(\widehat{g})] \ge \frac{3h^2 c_\mu}{2^{b+3}} M^{-b}.$$

Step 5: directional error is a Sharpe-ratio gap. Write

$$m := \langle g_\theta, \widehat{g}\rangle_{L^2(\nu)}, \qquad v := \|\widehat{g}\|^2_{L^2(\nu)}.$$

If $v > 0$, the payoff generated by $\widehat{g}$ satisfies

$$\mathbb{E}_\theta[R(\widehat{g})] = m, \qquad \mathbb{E}_\theta[R(\widehat{g})^2] = v,$$

and hence

$$\mathrm{SR}_\theta(\widehat{g})^2 = \frac{m^2}{v - m^2}.$$

Moreover,

$$D_\theta(\widehat{g}) = A_M - \frac{m^2}{v} = \frac{1}{1+\mathrm{SR}_\theta(\widehat{g})^2} - \frac{1}{1+(\mathrm{SR}^\star_\theta)^2}.$$

The same identity holds for $v = 0$ under the convention $\mathrm{SR}_\theta(\widehat{g}) = 0$. Cauchy–Schwarz implies $|\mathrm{SR}_\theta(\widehat{g})| \le \mathrm{SR}^\star_\theta$.

If $\mathrm{SR}_\theta(\widehat{g}) \ge 0$, then

$$\mathrm{SR}^\star_\theta - \mathrm{SR}_\theta(\widehat{g}) \ge \frac{D_\theta(\widehat{g})}{2\overline{S}}.$$

If $\mathrm{SR}_\theta(\widehat{g}) < 0$, then the gap is at least $\underline{S}$; since $D_\theta(\widehat{g}) \le A_M \le 1/4$, it is again bounded below by a fixed positive multiple of $D_\theta(\widehat{g})$. Therefore there is a constant $c_S > 0$, independent of M, T, θ, such that

$$\mathrm{SR}^\star_\theta - \mathrm{SR}_\theta(\widehat{g}) \ge c_S D_\theta(\widehat{g}).$$

Combining this inequality with Step 4 yields

$$2^{-M} \sum_\theta \mathbb{E}_\theta\big[\mathrm{SR}^\star_\theta - \mathrm{SR}_\theta(\widehat{g})\big] \ge c_1 M^{-b}$$

for a constant $c_1 > 0$. Since $M = \lceil T^{1/(b+1)} \rceil \le 2T^{1/(b+1)}$ for all sufficiently large T, we obtain

$$\sup_\theta \mathbb{E}_\theta\big[\mathrm{SR}^\star_\theta - \mathrm{SR}_\theta(\widehat{g})\big] \ge c_2 T^{-b/(b+1)}$$

for some $c_2 > 0$.

The argument allowed $\widehat{g}$ to range over all square-integrable estimators in the more informative augmented experiment, a larger class than managed portfolio rules based on the observed financial rows. Taking the infimum over all measurable portfolio-learning procedures proves

$$\inf_{\widetilde{W}_T} \sup_{\varepsilon \in \mathcal{E}} \mathbb{E}_\varepsilon \left[\mathrm{SR}^\star_{\varepsilon,K} - \mathrm{SR}_\varepsilon\Big(R^p_{t+1}(\widetilde{W}_T)\Big)\right] \ge c\, T^{-b/(b+1)}$$

for a constant $c > 0$ and all sufficiently large T. □

B. Tabelle

Table IV. Portfolio representations generated by different kernels

Representation	Kernel	Portfolio policy
Common exposure	$K(z, z') = 1$	$W(z)$ is constant
Linear	$K(z, z') = 1 + z'z'$	$W(z)$ is linear in characteristics
Finite features	$K(z, z') = \Phi(z)'\Phi(z')$	$W(z)$ lies in the span of the chosen features
Polynomial	$K(z, z') = (1 + z'z')^q$	$W(z)$ includes polynomial terms and interactions
Gaussian	$K(z, z') = \exp\left(-\frac{\|z - z'\|^2}{2\ell^2}\right)$	Flexible nonlinear $W(z)$
Matérn	$K_s(z, z') = c_{s,D}\, \rho_s(z, z')^{s-D/2} \times \mathcal{K}_{s-D/2}\big(\rho_s(z, z')\big)$	Flexible nonlinear $W(z)$ with smoothness controlled by s
Neural tangent	$K_{\mathrm{NTK}}(z, z') = \frac{\|z\|\,\|z'\|}{2\pi}\Big[\sin\theta + 2(\pi - \theta)\cos\theta\Big]$	Infinite-width one-hidden-layer ReLU policy

Notes: Here $\ell > 0$ denotes the kernel length scale. For the Matérn kernel,

$$\rho_s(z, z') = \sqrt{2s - D}\,\frac{\|z - z'\|}{\ell}, \qquad c_{s,D} = \frac{2^{1-(s-D/2)}}{\Gamma(s - D/2)}, \qquad s > \frac{D}{2},$$

and $\mathcal{K}_\alpha$ denotes the modified Bessel function of the second kind. Under standard regularity conditions, the Matérn RKHS is equivalent to a Sobolev space of smoothness s.
For the neural tangent kernel,

$$\theta = \arccos\left(\frac{z'z'}{\|z\|\,\|z'\|}\right),$$

and the displayed expression corresponds to a one-hidden-layer ReLU network without bias in the infinite-width NTK limit, up to the conventional overall normalization.
The kernel determines the class of characteristic-to-exposure mappings available to the portfolio manager. Portfolio returns use the common cross-sectional normalization in Equation (1). The common-exposure specification is an equal-exposure characteristic rule, not an unrestricted vector of static stock weights.

Table V relates the learning exponent to familiar spectral benchmarks. These are conditions on the managed-portfolio operator, not conclusions implied by the kernel name alone.

Table V. Benchmark portfolio-learning rates

Portfolio class	Managed-portfolio spectrum	Sharpe learning rate
Finite dimensional	Fixed finite rank	T^{-1}
Gaussian benchmark	Fast decay	Nearly T^{-1}
Sobolev–Matérn benchmark	$b = 2\widetilde{s}/D$	$T^{-2\widetilde{s}/(2\widetilde{s}+D)}$

Notes: Sharpe rates use Condition (E6). The finite-rank rate uses a fixed identifiable payoff space and root-T moment estimation. The fast-decay row is a separate i.i.d. or sufficiently fast-mixing benchmark, with possible logarithmic factors and an admissible regularization sequence; it does not follow by setting $b = \infty$ in Theorem 1. The Sobolev–Matérn row assumes the displayed *managed-payoff* spectral law and $\gamma > b$.

Population-optimal portfolio policy in a factor economy

Proposition. Consider the three-factor return model

$$R_{i,t+1} = \beta(Z_{i,t})^\top F_{t+1} + \varepsilon_{i,t+1}, \qquad i = 1, \dots, N,$$

where $\beta(z) \in \mathbb{R}^3$ and the characteristics $Z_{i,t}$ are measurable with respect to $\mathcal{F}_t$. Assume that

$$\begin{aligned} \mathbb{E}[F_{t+1} \mid \mathcal{F}_t] = \mu_F, \quad & \mathbb{E}[F_{t+1}F_{t+1}^\top \mid \mathcal{F}_t] = I_3, \\ \mathbb{E}[\varepsilon_{t+1} \mid \mathcal{F}_t] = 0, \quad & \mathbb{E}[\varepsilon_{t+1}\varepsilon_{t+1}^\top \mid \mathcal{F}_t] = \sigma_\varepsilon^2 I_N, \\ \mathbb{E}[F_{t+1}\varepsilon_{t+1}^\top \mid \mathcal{F}_t] = 0, & \qquad \sigma_\varepsilon^2 > 0. \end{aligned}$$

Suppose that the characteristics have a common marginal distribution P_Z, that $\beta_1, \beta_2, \beta_3 \in \mathcal{H}_K$, and that the factor loadings satisfy

$$\frac{1}{N}\sum_{i=1}^{N} \beta(Z_{i,t})\beta(Z_{i,t})^\top = gI_3, \qquad g = \frac{c_\beta}{3} > 0,$$

almost surely at every date. Then the population-optimal policy minimizing

$$Q(W) = \mathbb{E}\left[\left(1 - \frac{1}{N}\sum_{i=1}^{N} W(Z_{i,t})R_{i,t+1}\right)^2\right]$$

over $W \in \mathcal{H}_K$ is given, uniquely up to P_Z-almost-sure equality, by

$$W^\star(z) = \frac{\beta(z)^\top \mu_F}{g + \sigma_\varepsilon^2/N}.$$

Proof. For any $W \in \mathcal{H}_K$, define the portfolio factor exposure

$$b_t(W) = \frac{1}{N}\sum_{i=1}^{N} W(Z_{i,t})\beta(Z_{i,t}) \in \mathbb{R}^3.$$

Substituting the factor model into the portfolio payoff yields

$$R^p_{t+1}(W) = b_t(W)^\top F_{t+1} + \frac{1}{N}\sum_{i=1}^{N} W(Z_{i,t})\varepsilon_{i,t+1}.$$

Using the conditional moment assumptions and the law of iterated expectations, we obtain

$$\mathbb{E}[R^p_{t+1}(W) \mid \mathcal{F}_t] = b_t(W)^\top \mu_F$$

and

$$\mathbb{E}[(R^p_{t+1}(W))^2 \mid \mathcal{F}_t] = \|b_t(W)\|^2 + \frac{\sigma_\varepsilon^2}{N^2}\sum_{i=1}^{N} W(Z_{i,t})^2.$$

Consequently, using the common marginal distribution of the characteristics, the unconditional population loss becomes

$$Q(W) = 1 - 2\mu_F^\top \mathbb{E}[b_t(W)] + \mathbb{E}[\|b_t(W)\|^2] + \frac{\sigma_\varepsilon^2}{N}\mathbb{E}[W(Z)^2].$$

To compute its functional derivative, take an arbitrary direction $V \in \mathcal{H}_K$ and consider $W_h = W + hV$, with $h \in \mathbb{R}$. Since b_t is linear, we have $b_t(W + hV) = b_t(W) + hb_t(V)$. Substituting into the loss and expanding gives

$$Q(W + hV) = Q(W) + 2h\left[-\mu_F^\top \mathbb{E}[b_t(V)] + \mathbb{E}[b_t(W)^\top b_t(V)] + \frac{\sigma_\varepsilon^2}{N}\mathbb{E}[W(Z)V(Z)]\right] + h^2\left[\mathbb{E}[\|b_t(V)\|^2] + \frac{\sigma_\varepsilon^2}{N}\mathbb{E}[V(Z)^2]\right].$$

Therefore, the Gâteaux derivative of Q at W in the direction V is

$$\begin{aligned} DQ(W)[V] &= \frac{d}{dh}Q(W + hV)\Big|_{h=0} \\ &= -2\mu_F^\top \mathbb{E}[b_t(V)] + 2\mathbb{E}[b_t(W)^\top b_t(V)] + \frac{2\sigma_\varepsilon^2}{N}\mathbb{E}[W(Z)V(Z)]. \end{aligned}$$

A population minimizer must satisfy $DQ(W)[V] = 0$ for every $V \in \mathcal{H}_K$. Dividing by two and recalling that

$$b_t(V) = \frac{1}{N}\sum_{i=1}^{N} V(Z_{i,t})\beta(Z_{i,t}),$$

the functional first-order condition can be written as

$$0 = \mathbb{E}\left[\frac{1}{N}\sum_{i=1}^{N} V(Z_{i,t})\left(\beta(Z_{i,t})^\top [b_t(W) - \mu_F] + \frac{\sigma_\varepsilon^2}{N}W(Z_{i,t})\right)\right].$$

We solve this equation by imposing the stronger pointwise condition

$$\beta(Z_{i,t})^\top [b_t(W) - \mu_F] + \frac{\sigma_\varepsilon^2}{N}W(Z_{i,t}) = 0$$

for every i and almost every realization of the characteristics. This condition is sufficient, although not necessarily equivalent to the functional first-order condition, since it makes the expression inside the expectation vanish for every admissible direction V.

Isolating $W(Z_{i,t})$, we obtain

$$W(Z_{i,t}) = \frac{N}{\sigma_\varepsilon^2}\beta(Z_{i,t})^\top[\mu_F - b_t(W)].$$

Substituting this expression into the definition of $b_t(W)$ yields

$$\begin{aligned} b_t(W) &= \frac{1}{N}\sum_{i=1}^{N}\beta(Z_{i,t})W(Z_{i,t}) \\ &= \frac{1}{\sigma_\varepsilon^2}\left(\sum_{i=1}^{N}\beta(Z_{i,t})\beta(Z_{i,t})^\top\right)[\mu_F - b_t(W)]. \end{aligned}$$

By the cross-sectional balance condition, the matrix inside the parentheses equals NgI_3. Hence

$$b_t(W) = \frac{Ng}{\sigma_\varepsilon^2}[\mu_F - b_t(W)],$$

which implies

$$(Ng + \sigma_\varepsilon^2)b_t(W) = Ng\mu_F, \qquad b_t(W) = \frac{Ng}{Ng + \sigma_\varepsilon^2}\mu_F.$$

Substituting back into the expression for $W(Z_{i,t})$, we find

$$\begin{aligned} W(Z_{i,t}) &= \frac{N}{\sigma_\varepsilon^2}\beta(Z_{i,t})^\top\left[\mu_F - \frac{Ng}{Ng + \sigma_\varepsilon^2}\mu_F\right] \\ &= \frac{N}{\sigma_\varepsilon^2}\beta(Z_{i,t})^\top\frac{\sigma_\varepsilon^2}{Ng + \sigma_\varepsilon^2}\mu_F \\ &= \frac{\beta(Z_{i,t})^\top\mu_F}{g + \sigma_\varepsilon^2/N}. \end{aligned}$$

We have thus derived the function

$$W^\star(z) = \frac{\beta(z)^\top\mu_F}{g + \sigma_\varepsilon^2/N} = \frac{\beta(z)^\top\mu_F}{c_\beta/3 + \sigma_\varepsilon^2/N}.$$

Its functional form was not imposed in advance. Moreover, since $\beta_1, \beta_2, \beta_3 \in \mathcal{H}_K$, the resulting

$W^\star$ belongs to $\mathcal{H}_K$.

We now verify that this function satisfies the functional first-order condition. Using cross-sectional balance,

$$\begin{aligned} b_t(W^\star) &= \frac{1}{N}\sum_{i=1}^{N}\beta(Z_{i,t})W^\star(Z_{i,t}) \\ &= \frac{1}{g+\sigma_\varepsilon^2/N}\left(\frac{1}{N}\sum_{i=1}^{N}\beta(Z_{i,t})\beta(Z_{i,t})^\top\right)\mu_F \\ &= \frac{g}{g+\sigma_\varepsilon^2/N}\mu_F. \end{aligned}$$

Therefore, for every i and almost every realization of the characteristics,

$$\begin{aligned} &\beta(Z_{i,t})^\top[b_t(W^\star)-\mu_F] + \frac{\sigma_\varepsilon^2}{N}W^\star(Z_{i,t}) \\ &\quad = -\frac{\sigma_\varepsilon^2/N}{g+\sigma_\varepsilon^2/N}\beta(Z_{i,t})^\top\mu_F + \frac{\sigma_\varepsilon^2/N}{g+\sigma_\varepsilon^2/N}\beta(Z_{i,t})^\top\mu_F = 0. \end{aligned}$$

It follows that $DQ(W^\star)[V]=0$ for every $V\in\mathcal{H}_K$.

To establish global optimality, observe that the second functional variation is

$$D^2Q(W)[V,V] = 2\mathbb{E}[\|b_t(V)\|^2] + \frac{2\sigma_\varepsilon^2}{N}\mathbb{E}[V(Z)^2].$$

Both terms are nonnegative, and the second is strictly positive whenever V is nonzero in $L^2(P_Z)$. Thus Q is strictly convex up to P_Z-almost-sure equality, and the stationary point $W^\star$ is its unique population minimizer in this sense.

More explicitly, for any $W\in\mathcal{H}_K$, let $\Delta W = W - W^\star$. Since the first functional variation vanishes at $W^\star$, the quadratic expansion gives the exact excess-loss identity

$$Q(W)-Q(W^\star) = \mathbb{E}[\|b_t(\Delta W)\|^2] + \frac{\sigma_\varepsilon^2}{N}\mathbb{E}[\Delta W(Z)^2] \geq 0.$$

Finally, the solution also satisfies the conditional pricing restriction. Let $B_t\in\mathbb{R}^{N\times 3}$ contain the

factor loadings as its rows and define

$$w_t^\star = \frac{1}{N}(W^\star(Z_{1,t}), \dots, W^\star(Z_{N,t}))^\top, \qquad v_t = \frac{1}{N}(V(Z_{1,t}), \dots, V(Z_{N,t}))^\top.$$

The pointwise condition established above is equivalent to

$$(B_t B_t^\top + \sigma_\varepsilon^2 I_N) w_t^\star = B_t \mu_F.$$

Consequently, for every $V \in \mathcal{H}_K$,

$$\begin{aligned}&\mathbb{E}[(1 - R_{t+1}^p(W^\star))R_{t+1}^p(V) \mid \mathcal{F}_t] \\ &\qquad = v_t^\top B_t \mu_F - v_t^\top (B_t B_t^\top + \sigma_\varepsilon^2 I_N) w_t^\star = 0.\end{aligned}$$

Hence the population-optimal policy also satisfies Condition E4. □

For the conditional Gaussian specification

$$F_{t+1} \sim \mathcal{N}(\mu_F, I_3 - \mu_F \mu_F^\top), \qquad \varepsilon_{t+1} \sim \mathcal{N}(0, \sigma_\varepsilon^2 I_N),$$

with conditionally independent factors and idiosyncratic innovations, and $\|\beta(z)\|^2 = c_\beta$. Then

$$\mathbb{E}[R_{i,t+1}^2 \mid \mathcal{F}_t] = \|\beta(Z_{i,t})\|^2 + \sigma_\varepsilon^2 = c_\beta + \sigma_\varepsilon^2.$$

Proposition 4 therefore verifies Condition (E2) with $B_X^2 = 8\kappa_K(c_\beta + \sigma_\varepsilon^2)$ and $K_4 = 3$. For a normalized Matérn kernel, $\kappa_K = 1$.

Proposition 5 (Eigenvalue decay of the managed-return covariance operator)**.** *Consider the factor economy*

$$R_{i,t+1} = \beta(Z_{i,t})^\top F_{t+1} + \varepsilon_{i,t+1}, \qquad i = 1, \dots, N,$$

where

$$\mathbb{E}[F_{t+1}F_{t+1}^{\top} \mid \mathcal{F}_t] = I_3, \qquad \mathbb{E}[\varepsilon_{t+1}\varepsilon_{t+1}^{\top} \mid \mathcal{F}_t] = \sigma_\varepsilon^2 I_N, \qquad \mathbb{E}[F_{t+1}\varepsilon_{t+1}^{\top} \mid \mathcal{F}_t] = 0,$$

with $\sigma_\varepsilon^2 > 0$*. Assume that the characteristics have a common uniform marginal distribution* P_Z *on* $\mathcal{Z} = [-1, 1]^D$ *and that the factor loadings satisfy*

$$\frac{1}{N}\sum_{i=1}^{N} \beta(Z_{i,t})\beta(Z_{i,t})^{\top} = gI_3, \qquad g > 0,$$

almost surely at every date. Let K *be a Matérn kernel with smoothness parameter* $\nu > 0$ *and fixed positive length scale, and define*

$$X_t = \frac{1}{N}\sum_{i=1}^{N} R_{i,t+1}K(\cdot, Z_{i,t}), \qquad \Sigma_{\mathcal{H}_K,\varepsilon} = \mathbb{E}_\varepsilon[X_t \otimes X_t].$$

Then, for every fixed economy with finite N*, the positive eigenvalues* $\mu_{j,\varepsilon}$ *of* $\Sigma_{\mathcal{H}_K,\varepsilon}$ *satisfy*

$$c_\mu j^{-b} \le \mu_{j,\varepsilon} \le C_\mu j^{-b}, \qquad j \ge 1,$$

where $b = 1 + 2\nu/D$ *and* $0 < c_\mu \le C_\mu < \infty$*. In particular, for* $D = 6$ *and* $\nu = 3/2$*, we obtain* $b = 3/2$*.*

Proof. Consider an arbitrary function $f \in \mathcal{H}_K$. By the definition of the covariance operator and the reproducing property of the kernel,

$$\begin{aligned}\langle f, \Sigma_{\mathcal{H}_K,\varepsilon} f\rangle_{\mathcal{H}_K} &= \mathbb{E}_\varepsilon[\langle f, X_t\rangle_{\mathcal{H}_K}^2] \\ &= \mathbb{E}_\varepsilon\left[\left(\frac{1}{N}\sum_{i=1}^{N} f(Z_{i,t})R_{i,t+1}\right)^2\right].\end{aligned}$$

Substituting the factor model, we obtain

$$\frac{1}{N}\sum_{i=1}^{N} f(Z_{i,t})R_{i,t+1} = \left(\frac{1}{N}\sum_{i=1}^{N} f(Z_{i,t})\beta(Z_{i,t})\right)^{\top} F_{t+1} + \frac{1}{N}\sum_{i=1}^{N} f(Z_{i,t})\varepsilon_{i,t+1}.$$

Since the characteristics are $\mathcal{F}_t$-measurable, the conditional second-moment assumptions imply

$$\begin{aligned}\mathbb{E}_{\varepsilon}&\left[\left(\frac{1}{N}\sum_{i=1}^{N} f(Z_{i,t})R_{i,t+1}\right)^2 \middle| \mathcal{F}_t\right] \\ &= \left\|\frac{1}{N}\sum_{i=1}^{N} f(Z_{i,t})\beta(Z_{i,t})\right\|^2 + \frac{\sigma_{\varepsilon}^2}{N^2}\sum_{i=1}^{N} f(Z_{i,t})^2.\end{aligned}$$

Taking unconditional expectations and using the common marginal distribution of the characteristics gives

$$\langle f, \Sigma_{\mathcal{H}_K,\varepsilon} f\rangle_{\mathcal{H}_K} = \mathbb{E}_{\varepsilon}\left[\left\|\frac{1}{N}\sum_{i=1}^{N} f(Z_{i,t})\beta(Z_{i,t})\right\|^2\right] + \frac{\sigma_{\varepsilon}^2}{N}\mathbb{E}_{\varepsilon}[f(Z)^2].$$

Define the kernel covariance operator

$$T_K = \mathbb{E}_{\varepsilon}[K(\cdot, Z) \otimes K(\cdot, Z)],$$

where $Z \sim P_Z$. By the reproducing property,

$$\langle f, T_K f\rangle_{\mathcal{H}_K} = \mathbb{E}_{\varepsilon}[f(Z)^2].$$

Since the first term in the preceding expression for $\langle f, \Sigma_{\mathcal{H}_K,\varepsilon} f\rangle_{\mathcal{H}_K}$ is nonnegative, we immediately obtain

$$\langle f, \Sigma_{\mathcal{H}_K,\varepsilon} f\rangle_{\mathcal{H}_K} \geq \frac{\sigma_{\varepsilon}^2}{N}\langle f, T_K f\rangle_{\mathcal{H}_K}.$$

To establish the upper bound, let $B_t \in \mathbb{R}^{N\times 3}$ denote the matrix whose i-th row is $\beta(Z_{i,t})^{\top}$, and

write $f_t = (f(Z_{1,t}), \dots, f(Z_{N,t}))^\top$. The cross-sectional balance condition implies

$$B_t^\top B_t = NgI_3.$$

Consequently, the largest eigenvalue of $B_t B_t^\top$ is Ng, and hence

$$\left\| \frac{1}{N} \sum_{i=1}^{N} f(Z_{i,t})\beta(Z_{i,t}) \right\|^2 = \frac{1}{N^2} f_t^\top B_t B_t^\top f_t$$
$$\leq \frac{g}{N}\|f_t\|^2 = \frac{g}{N} \sum_{i=1}^{N} f(Z_{i,t})^2.$$

Taking expectations and using the common marginal law,

$$\mathbb{E}_\varepsilon \left[\left\| \frac{1}{N} \sum_{i=1}^{N} f(Z_{i,t})\beta(Z_{i,t}) \right\|^2 \right] \leq g\mathbb{E}_\varepsilon[f(Z)^2].$$

Combining this inequality with the idiosyncratic component yields

$$\langle f, \Sigma_{\mathcal{H}_K,\varepsilon} f\rangle_{\mathcal{H}_K} \leq \left(g + \frac{\sigma_\varepsilon^2}{N}\right) \langle f, T_K f\rangle_{\mathcal{H}_K}.$$

Therefore, for every $f \in \mathcal{H}_K$,

$$\frac{\sigma_\varepsilon^2}{N}\langle f, T_K f\rangle_{\mathcal{H}_K} \leq \langle f, \Sigma_{\mathcal{H}_K,\varepsilon} f\rangle_{\mathcal{H}_K} \leq \left(g + \frac{\sigma_\varepsilon^2}{N}\right) \langle f, T_K f\rangle_{\mathcal{H}_K}.$$

Equivalently, in the Loewner ordering of self-adjoint operators,

$$\frac{\sigma_\varepsilon^2}{N} T_K \preceq \Sigma_{\mathcal{H}_K,\varepsilon} \preceq \left(g + \frac{\sigma_\varepsilon^2}{N}\right) T_K.$$

Let $\lambda_j(T_K)$ denote the eigenvalues of T_K, arranged in nonincreasing order. Since both operators

are positive, self-adjoint and compact, the min–max principle implies

$$\frac{\sigma_\varepsilon^2}{N}\lambda_j(T_K) \le \mu_{j,\varepsilon} \le \left(g + \frac{\sigma_\varepsilon^2}{N}\right)\lambda_j(T_K), \qquad j \ge 1.$$

For a Matérn kernel with smoothness parameter ν on the bounded Lipschitz domain $\mathcal{Z} = [-1,1]^D$, equipped with a uniform probability measure, the standard spectral asymptotics of the associated kernel integral operator imply that there exist constants $0 < c_K \le C_K < \infty$ such that

$$c_K j^{-(1+2\nu/D)} \le \lambda_j(T_K) \le C_K j^{-(1+2\nu/D)}, \qquad j \ge 1.$$

The nonzero eigenvalues of the kernel integral operator on $L^2(P_Z)$ coincide with those of T_K on $\mathcal{H}_K$. Therefore, combining the preceding inequalities, we obtain

$$\frac{\sigma_\varepsilon^2}{N}c_K j^{-(1+2\nu/D)} \le \mu_{j,\varepsilon} \le \left(g + \frac{\sigma_\varepsilon^2}{N}\right)C_K j^{-(1+2\nu/D)}.$$

Thus the required polynomial eigenvalue decay holds with

$$b = 1 + \frac{2\nu}{D}, \qquad c_\mu = \frac{\sigma_\varepsilon^2}{N}c_K, \qquad C_\mu = \left(g + \frac{\sigma_\varepsilon^2}{N}\right)C_K.$$

In the baseline specification, $\nu = 3/2$ and $D = 6$, so that

$$b = 1 + \frac{2(3/2)}{6} = \frac{3}{2}.$$

Hence,

$$c_\mu j^{-3/2} \le \mu_{j,\varepsilon} \le C_\mu j^{-3/2}, \qquad j \ge 1,$$

which establishes the claimed eigenvalue decay. □

Proposition 6 (Uniform bounds for the optimal Sharpe ratio)**.** *Consider the factor economy de-*

scribed above, with $N = 3G \geq 3$, $g = c_\beta/3 > 0$, $\sigma_\varepsilon^2 > 0$, *and* $0 < \|\mu_F\|^2 < 1$. *Let*

$$W^\star(z) = \frac{\beta(z)^\top \mu_F}{g + \sigma_\varepsilon^2/N}$$

denote the population-optimal portfolio policy. Then there exist constants $0 < SR_{\min} \leq SR_{\max} < \infty$, *independent of* N, *such that*

$$SR_{\min} \leq SR(W^\star) \leq SR_{\max}.$$

Proof. Recall that the portfolio return associated with a policy W is

$$R_{t+1}^p(W) = \frac{1}{N} \sum_{i=1}^{N} W(Z_{i,t}) R_{i,t+1}.$$

Its Sharpe ratio is defined as

$$SR(W) = \frac{\mathbb{E}[R_{t+1}^p(W)]}{\sqrt{\mathrm{Var}(R_{t+1}^p(W))}}.$$

For the population-optimal policy, the cross-sectional balance condition implies

$$b_t(W^\star) = \frac{1}{N} \sum_{i=1}^{N} W^\star(Z_{i,t}) \beta(Z_{i,t}) = \frac{g}{g + \sigma_\varepsilon^2/N} \mu_F.$$

Consequently,

$$\mathbb{E}[R_{t+1}^p(W^\star)] = \mu_F^\top b_t(W^\star) = \frac{g\|\mu_F\|^2}{g + \sigma_\varepsilon^2/N}.$$

Using the conditional second-moment assumptions of the factor model, we also have

$$\begin{aligned}
\mathbb{E}[(R_{t+1}^p(W^\star))^2] &= \mathbb{E}[\|b_t(W^\star)\|^2] + \frac{\sigma_\varepsilon^2}{N} \mathbb{E}[W^\star(Z)^2] \\
&= \frac{g^2\|\mu_F\|^2}{(g + \sigma_\varepsilon^2/N)^2} + \frac{\sigma_\varepsilon^2}{N} \frac{g\|\mu_F\|^2}{(g + \sigma_\varepsilon^2/N)^2} \\
&= \frac{g\|\mu_F\|^2}{g + \sigma_\varepsilon^2/N},
\end{aligned}$$

where we used $\mathbb{E}[\beta(Z)\beta(Z)^\top] = gI_3$.

Define

$$q_N = \frac{g\|\mu_F\|^2}{g + \sigma_\varepsilon^2/N}.$$

The preceding identities imply

$$\mathbb{E}[R_{t+1}^p(W^\star)] = \mathbb{E}[(R_{t+1}^p(W^\star))^2] = q_N,$$

and therefore

$$\mathrm{Var}(R_{t+1}^p(W^\star)) = q_N - q_N^2.$$

It follows that

$$SR(W^\star) = \frac{q_N}{\sqrt{q_N - q_N^2}} = \sqrt{\frac{q_N}{1 - q_N}}.$$

Since $N \geq 3$, we have

$$0 < \frac{\sigma_\varepsilon^2}{N} \leq \frac{\sigma_\varepsilon^2}{3}.$$

Together with $g > 0$ and $0 < \|\mu_F\|^2 < 1$, this yields

$$0 < \frac{g\|\mu_F\|^2}{g + \sigma_\varepsilon^2/3} \leq q_N \leq \|\mu_F\|^2 < 1.$$

Since the function $q \mapsto \sqrt{q/(1-q)}$ is strictly increasing on $(0, 1)$, we obtain

$$\sqrt{\frac{g\|\mu_F\|^2}{g + \sigma_\varepsilon^2/3 - g\|\mu_F\|^2}} \leq SR(W^\star) \leq \sqrt{\frac{\|\mu_F\|^2}{1 - \|\mu_F\|^2}}.$$

Thus, defining

$$SR_{\min} = \sqrt{\frac{g\|\mu_F\|^2}{g + \sigma_\varepsilon^2/3 - g\|\mu_F\|^2}}, \qquad SR_{\max} = \sqrt{\frac{\|\mu_F\|^2}{1 - \|\mu_F\|^2}},$$

we conclude that

$$0 < SR_{\min} \leq SR(W^{\star}) \leq SR_{\max} < \infty.$$

Both constants depend only on the fixed model parameters g, μ_F, and σ_ε^2, and are independent of the number of assets N. This establishes the required uniform Sharpe-ratio bounds. □